A Trichotomy for Monadic Theories of Linear Recurrence Sequences

Toghrul Karimov, Joris Nieuwveld, Joël Ouaknine

Max Planck Institute for Software Systems, Saarland Informatics Campus, Saarbrücken, 66123, Germany

Abstract

Expansions of the monadic second-order (MSO) theory of the structure $\langle \mathbb{N}; < \rangle$ have been a fertile and active area of research ever since the publication of the seminal papers of Büchi and Elgot & Rabin on the subject in the 1960s. In the present paper, we establish decidability of the MSO theory of $\langle \mathbb{N}; <, P \rangle$, where P ranges over a large class of unary predicates, specifically, sets of non-negative values assumed by certain integer linear recurrence sequences. One of our key technical tools is the novel concept of *(effective) prodisjunctivity*, which we expect may also find independent applications further afield.

Keywords: Monadic second-order logic, linear recurrence sequences, decidability, Baker's theorem

1. Introduction

The monadic second-order (MSO) theory of the structure $\langle \mathbb{N}; < \rangle$ has been a foundational pillar of the field of automated verification, and more generally, the area of logic in computer science, for many decades. Arguably the most important paper on the topic is due to Büchi [1], who in the early 1960s established decidability of this theory through a deep and fecund connection between logic and automata theory.

Shortly thereafter, in yet another seminal piece of work, Elgot and Rabin [2] devised the *contraction method* to establish decidability of expansions of this base theory by various "arithmetic" unary predicates $P \subseteq \mathbb{N}$. In particular, they proved decidability of the MSO theory of $\langle \mathbb{N}; <, P \rangle$, where P could for example be taken to be the set $2^{\mathbb{N}}$ of powers of 2, or the set Sq of perfect squares, or the set Fac of factorial numbers, and so on.

Much progress followed in the ensuing decades regarding decidability of MSO theories of expansions of $\langle \mathbb{N}; < \rangle$. Carton and Thomas [3] substantially refined Elgot and Rabin's automata-theoretic techniques into the notion of *effective profinitely ultimate periodicity*, which they used to show decidability of the MSO theory of $\langle \mathbb{N}; <, P \rangle$, for any P whose characteristic word $u \in \{0,1\}^\omega$ (which has 1 at the nth position if and only if $n \in P$) is *morphic*. Examples of morphic words include the Fibonacci word, the Tribonacci word, the Thue-Morse word, as well as the characteristic word of $\{p(n)a^n \colon n \in \mathbb{N}\} \cap \mathbb{N}$ for any polynomial $p \in \mathbb{Z}[X]$ and integer a. The theory of profinitely ultimately periodic words was further developed by Rabinovich and Thomas [4, 5], and its application to morphic words was recently generalised to S-adic words by Berthé *et al.* [6].

Semënov, on the other hand, (i) introduced the notion of *effective almost periodicity* [7], and (ii) gave a strong characterisation of predicates P for which the MSO theory $\langle \mathbb{N}; <, P \rangle$ is decidable [8]. His ideas have been used to establish many novel MSO decidability results. For example, Bateman, Jockusch, and Woods showed, using the characterisation of [8], that the MSO theory of $\langle \mathbb{N}; <, \{p \colon p \text{ is prime}\} \rangle$ is decidable assuming (the linear case of) Schinzel's Hypothesis H [9]. Almost periodic words, on the other hand, underlie a string of results ranging from novel algorithms for model checking *linear dynamical systems* [10] to conditional decidability (assuming Schanuel's conjecture) of the MSO theory of $\langle \mathbb{N}; <, a_1^{\mathbb{N}}, \ldots, a_d^{\mathbb{N}} \rangle$, where $a_i^{\mathbb{N}} = \{a_i^n \colon n \in \mathbb{N}\}$, for any positive integers $a_1, \ldots, a_d$ [11]. The latter work by Berthé *et al.* shows, using tools from number theory, symbolic dynamics, and word combinatorics, how to decide the MSO theory of *multiple* predicates (i.e., the MSO theory of $\langle \mathbb{N}; <, P_1, \ldots, P_d \rangle$) of arithmetic origin, which had remained an open problem since the breakthrough work of Elgot and Rabin mentioned earlier.

Let us also mention that for the vast majority of functions $f \colon \mathbb{N} \to \mathbb{N}$ not already definable in $\langle \mathbb{N}; < \rangle$, the MSO theory of $\langle \mathbb{N}; <, f \rangle$ is undecidable [12]. Hence the bulk of research into decidability of MSO theories of expansions of $\langle \mathbb{N}; < \rangle$ has focussed on unary predicates. At the time of writing, we do not know of any "natural" predicate P for which the MSO theory of $\langle \mathbb{N}; <, P \rangle$ is undecidable: for most P of arithmetic origin, decidability simply remains open. However, see [13, Sec. 3.5] for an example of a predicate P for which already the *first-order* theory of $\langle \mathbb{N}; <, P \rangle$ is undecidable.

In the present paper, we extend the research into decidable MSO theories by considering predicates derived from *integer linear recurrence sequences*

(LRS). Specifically, we consider *simple* LRS with at most three *dominant characteristic roots.*[1] The number of dominant roots of an LRS, as well as whether it is simple or not, are the two best-known notions used to quantify the complexity of a given LRS. For P the set of non-negative values of an LRS having a single dominant root, such as the set of Fibonacci numbers, the decidability of the MSO theory of the structure $\langle \mathbb{N}; <, P\rangle$ is readily established via Elgot and Rabin's contraction method;[2] see also [11] in which expansions of $\langle \mathbb{N}; <\rangle$ by adjoining multiple predicates obtained from such LRS are thoroughly investigated. However, the situation with LRS having more than a single dominant characteristic root, which we address in this paper, is hitherto completely uncharted. Our main contribution is as follows:

Theorem 1.1. *Let $P = \{u_n \colon n \in \mathbb{N}\} \cap \mathbb{N}$ for $\langle u_n\rangle_{n=0}^{\infty}$ a non-degenerate, simple, integer-valued linear recurrence sequence with at most three dominant roots. Then the MSO theory of the structure $\langle \mathbb{N}; <, P\rangle$ is decidable.*

One notices that here we are discarding all the negative values of our LRS, which is, of course, necessary since we are working over domain $\mathbb{N}$ rather than $\mathbb{Z}$. At a technical level, working over $\mathbb{N}$ allows us to use classical automata theory, rather than the theory of automata on bi-infinite words. The restriction to $\mathbb{N}$, however, is entirely benign in view of the following:

Corollary 1.2. *Let $\overline{P} = \{u_n \colon n \in \mathbb{N}\}$ for $\langle u_n\rangle_{n=0}^{\infty}$ a non-degenerate, simple, integer-valued linear recurrence sequence with at most three dominant roots. Then the MSO theory of the structure $\langle \mathbb{Z}; 0, <, \overline{P}\rangle$ is decidable.*

This result is straightforwardly obtained from Thm. 1.1 via an application of the celebrated "composition method" of Shelah in model theory [14]. In the case at hand, one can directly invoke, e.g., [15, Cor. 6], since the structure $\langle \mathbb{Z}; 0, <, \overline{P}\rangle$ is isomorphic to the *ordered sum* $\langle \mathbb{Z} - \mathbb{N}; <, P^-\rangle + \langle \mathbb{N}; <, P\rangle$ where P is as in Thm. 1.1 and $P^- = \{u_n \colon n \in \mathbb{N}\} \cap (\mathbb{Z} - \mathbb{N})$. (We have that the MSO theory of $\langle \mathbb{Z} - \mathbb{N}; <, P^-\rangle$ is decidable if and only if the MSO theory of $\langle \mathbb{N}; <, \{-u_n \colon n \in \mathbb{N}\} \cap \mathbb{N}\rangle$ is decidable, which is the case by Thm. 1.1.)

[1] These notions are properly defined in Sec. 2.4.

[2] Simple LRS having a single positive dominant root are particularly well-behaved: they are either constant, or effectively ultimately monotonically increasing (or decreasing), effectively *procyclic*, and *effectively sparse* (see [11] for precise definitions of these notions and a proper account of these facts). In the case of a negative dominant root, essentially the same behaviour is observed among the non-negative terms.

That is, MSO sentences over $\langle \mathbb{Z}; <, \overline{P} \rangle$ can be faithfully decomposed into component subformulas dealing exclusively with either non-negative or negative values of our LRS, with the truth value of the original sentence obtained by appropriately piecing together truth values of each of the sub-sentences within their respective structures.

Let us discuss our approach to proving Thm. 1.1. The main technical tool is the notion of *effective prodisjunctivity* (defined below), which was introduced in our earlier work [16]. Metaphorically speaking, prodisjunctivity is an analogue of the idea that "everything that has positive probability will happen infinitely often". In [16], the authors show that (i) for a simple, non-degenerate, integer-valued LRS $\langle u_n \rangle_{n=0}^{\infty}$ that has exactly two dominant roots, the predicate $P = \{u_n \colon n \in \mathbb{N}\} \cap \mathbb{N}$ is effectively prodisjunctive, and (ii) for such P, the MSO theory of $\langle \mathbb{N}; <, P \rangle$ is decidable. When considering (simple, non-degenerate, integer-valued) LRS with at most three dominant roots, we arrive at the following trichotomy: the corresponding predicate P is either finite, or almost periodic (in which case we can apply the aforementioned results of Semënov), or prodisjunctive.

We sketch out these three cases. Let $\langle u_n \rangle_{n=0}^{\infty}$ be an LRS as in Thm. 1.1.

The finite case

Our first case consists of LRS for which P is finite. One such LRS is given by the initial values $(u_0, u_1, u_2) = (0, -3, -15)$ and the linear recurrence

$$u_{n+3} = 3u_{n+2} - 6u_{n+1} + 8u_n \,. \tag{1}$$

Then, setting $\lambda = \frac{1+\sqrt{-15}}{2}$, we have that $u_n = \lambda^n + \overline{\lambda}^n - 2 \cdot 2^n$, and as $|\lambda| = 2$, $|\lambda^n + \overline{\lambda}^n| < 2 \cdot 2^n$, which implies that $u_n \leq 0$. Hence, as $u_0 = 0$, we have that $P = \{u_n \colon n \in \mathbb{N}\} \cap \mathbb{N} = \{0\}$. Thus, the MSO theory of the structure $\langle \mathbb{N}; <, P \rangle$ is decidable as P can be defined in $\langle \mathbb{N}; < \rangle$. In general, when P is finite, we can effectively determine all $n \in P$, and establishing decidability of the MSO theory of $\langle \mathbb{N}; <, P \rangle$ is straightforward.

The almost periodic case

When P is infinite, we can employ the Elgot-Rabin contraction method by enumerating P: Let p_{m-1} be the mth smallest element in P. As $p_{m+1} - p_m$ diverges to infinity in a relatively controlled way, following Elgot's and Rabin's footsteps, we only need to understand $\langle p_m \bmod M \rangle_{m=0}^{\infty}$ for all $M \geq 1$. Here, by "understand" we mean that we can decide whether a Büchi automaton

with alphabet $\{0, \ldots, M\}$ accepts $\langle p_m \bmod M\rangle_{m=0}^{\infty}$. Preferring deterministic automata, we work with the equally expressive class of deterministic Muller automata. We refer to this acceptance problem as $\mathrm{Acc}_{\langle p_m \bmod M\rangle_{m=0}^{\infty}}$.

We now encounter a complication: Although $\langle u_n \bmod M\rangle_{n=0}^{\infty}$ is ultimately periodic for all M, $\langle p_m\rangle_{m=0}^{\infty}$ lacks such a clear structure. In fact, discarding all the negative values of $\langle u_n \bmod M\rangle_{n=0}^{\infty}$ and sorting out the remaining ones in increasing order fundamentally alters the nature of the sequence. However, it turns out that exactly one of two different types of structure emerge.

To introduce the first structure, we move to an example. Take the LRS defined by the initial values $(u_0, u_1, u_2) = (11, 16, 14)$ and linear recurrence (1). Then $u_n = 2\lambda^n + 2\overline{\lambda}^n + 7 \cdot 2^n$ and the first few values of $\langle u_n\rangle_{n=0}^{\infty}$ are

$$11, 16, 14, 34, 146, 346, 434, 394, 1346, 5146, 10514, 11434, 12386, 52666, \ldots.$$

Thus $\langle u_n\rangle_{n=0}^{\infty}$ is generally increasing, with some exceptions: we see that $u_{n+1} < u_n$ for some n. However, elements cannot be sorted far away from their original position as $u_n > u_m$ whenever $n > m + 1$. Thus, we can turn $\langle u_n\rangle_{n=0}^{\infty}$ into $\langle p_m\rangle_{m=0}^{\infty}$ by "swapping" an infinite number of consecutive values of $\langle u_n\rangle_{n=0}^{\infty}$. Let $M = 3$. Then

$$(u_n \bmod M) = \begin{cases} 2 & \text{if } n \equiv 0 \pmod 2 \\ 1 & \text{if } n \equiv 1 \pmod 2. \end{cases}$$

However, because of our "swaps" when sorting the corresponding sequence $1, 2, 1, 2, \ldots$, the first few values of $\langle p_m \bmod 3\rangle_{m=0}^{\infty}$ are

$$2, 2, 1, 1, 2, 1, 1, 2, 2, 1, 2, 1, 2, 1, 2, 2, 1, 1, 2, 1, 1, 2, 2, 1, 2, 2, 1, 1, 2, 1, 2, 1, \ldots. \tag{2}$$

This sequence is not ultimately periodic, but the kind of patterns that appear are severely limited and occur frequently. For example, the same number never appears thrice in a row, and the pattern $2, 2$ appears in every subsequence of length 8. Formally speaking, the sequence $\langle p_m \bmod M\rangle_{m=0}^{\infty}$ conforms to Semënov's notion of (effective) almost periodicity, which we formally define in Sec. 2.2. Therefore, by Semënov's results, $\mathrm{Acc}_{\langle p_m \bmod 3\rangle_{m=0}^{\infty}}$ is decidable. More generally, we have the following.

Theorem 1.3. *Let $\langle u_n\rangle_{n=0}^{\infty}$ be a simple, non-degenerate LRS with two or three dominant roots such that P is infinite and there is a number D such that $u_n > u_m$ whenever $n > m + D$. Then, for any $M \geq 1$, the sequence $\langle u_n \bmod M\rangle_{m=0}^{\infty}$ is effectively almost periodic.*

The prodisjunctive case

Not all LRS $\boldsymbol{u}$ enjoy the property introduced in Thm. 1.3 to the effect that there is a number $D \geq 1$ such that $u_n > u_m$ whenever $n > m + D$. A counterexample such an LRS is given by the initial values $(u_0, u_1, u_2) = (2, 1, -7)$ and linear recurrence (1). Then, $u_n = 2\lambda^n + 2\overline{\lambda}^n$ and its first few terms are

$$2, 1, -7, -11, 17, 61, -7, -251, -223, 781, 1673, -1451, -8143, -2339, \ldots .$$

Writing $\theta = \arg(\lambda)$, we have that $u_n = 2\cos(\theta n)2^n$. As $\langle \theta n \rangle_{n=0}^{\infty}$ is dense in $\mathbb{R}/(2\pi\mathbb{Z})$, the number $\cos(\theta n)$ can be arbitrarily close to 0. Hence, for every $D \geq 0$, there are m, n such that $m < n + D$ and $u_n < u_m$.

Owing to this property, the enumeration $\langle p_m \bmod M \rangle_{m=0}^{\infty}$ is much more chaotic than in the almost periodic case. Taking $M = 3$, $\langle p_m \bmod 3 \rangle_{m=0}^{\infty}$ is

$$1, 2, 2, 1, 1, 2, 2, 1, 2, 2, 1, 1, 1, 2, 2, 1, 1, 1, 2, 1, 2, 2, 2, 1, 1, 2, 2, 1, 1, 2, 1, 1, \ldots .$$

Already, we see a larger variety of subsequences than in (2) as a symbol can appear three times in a row. In fact, every finite sequence of 1s and 2s appears. Let us be more precise and introduce the notion of *(effective) prodisjunctivity*, as follows.

Given a finite alphabet Σ together with a subset $S \subseteq \Sigma$, we say that an infinite sequence in Σ^{ω} is *disjunctive relative to* S if every finite word over S appears infinitely often as a factor in the sequence. We now have:

Definition 1.4. *Let $\langle p_m \rangle_{m=0}^{\infty}$ be an infinite integer-valued sequence. For any integer $M \geq 1$, let S_M be the set of residue classes modulo M that appear infinitely often in $\langle p_m \rangle_{m=0}^{\infty}$:*

$$S_M = \left\{ s \in \{0, \ldots, M-1\} \colon \exists^{\infty} m \in \mathbb{N} \colon p_m \equiv s \pmod{M} \right\}.$$

We say that $\langle p_m \rangle_{n=0}^{\infty}$ is ***prodisjunctive*** *if, for all $M \geq 1$, the sequence of residues $\langle p_m \bmod M \rangle_{m=0}^{\infty}$ is disjunctive relative to S_M. For* ***effective prodisjunctivity****, we require in addition that the sequence $\langle p_m \bmod M \rangle_{m=0}^{\infty}$ be computable and that, for each M, the set S_M, together with the smallest index threshold N_M beyond which all residue classes of the sequence lie in S_M (i.e. $p_m \bmod M \in S_M$ for all $m \geq N_M$), also be computable.*

Prodisjunctivity is itself predicated on the notion of disjunctivity, which has already been successfully applied to the decidability of logical theories

(under the alternative terminology of "weak normality") by Berthé *et al.* [11]. In particular, the authors of *op. cit.* establish decidability of the MSO theory of $\langle \mathbb{N}; <, 2^{\mathbb{N}}, \mathsf{Sq} \rangle$ under the assumption that the binary expansion of $\sqrt{2}$ is disjunctive, i.e., contains every finite bit pattern as a factor infinitely often.

The hypothesis that $\sqrt{2}$ is disjunctive in base 2 is widely expected by number theorists to be true, but remains a major unsolved problem. Irrational algebraic numbers, along with most known transcendental numbers such as e and π, are in fact believed to satisfy a stronger property, that of being *normal* in every integer base: a real number α is normal in base b if every finite word $w \in \{0, \ldots, b-1\}^*$ appears with frequency $b^{-|w|}$ in the base-b expansion of α. And although Borel [17] showed over a century ago that non-normal real numbers have null Lebesgue measure, establishing normality (or even disjunctivity) of everyday irrational numbers has remained fiercely elusive; see [18, 19, 20, 21] for more detailed accounts of results, research, and open problems in this area.

Let us now discuss how prodisjunctivity relates to Thm. 1.1.

Theorem 1.5. *Let $\langle p_m \rangle_{m=0}^{\infty}$ be as above, i.e., the strictly increasing sequence of positive terms of some non-degenerate, simple, integer-valued LRS having two or three dominant roots such that for all $D \geq 0$ there exists $n > m + D$ such that $0 \leq u_n < u_m$. Then $\langle p_m \rangle_{m=0}^{\infty}$ is effectively prodisjunctive.*

In the case at hand, our example sequence is disjunctive over $S_3 = \{1, 2\}$. The factors $\langle 1, 1, 1 \rangle$ and $\langle 2, 2, 2 \rangle$ appear frequently, at indices 10 and 15 for the first time, respectively, and both have a density of roughly one in 23. However, four consecutive identical symbols are rare in $\langle p_m \bmod 3 \rangle_{m=0}^{\infty}$: $\langle 1, 1, 1, 1 \rangle$ occurs for the first time at $m = 11233$, 44162, and 130919. Hence, prodisjunctive sequences of the form $\langle p_m \bmod M \rangle_{m=0}^{\infty}$ are typically not normal: Otherwise, for example, $\langle 1, 1, 1 \rangle$ would need to occur with frequency 2^{-3} and $\langle 1, 1, 1, 1 \rangle$ with frequency 2^{-4}, both of which are demonstrably false.

Theorems 1.3 and 1.5 are the key technical devices underpinning our main result, Thm. 1.1. They are proven using Baker's celebrated theorem on linear forms in logarithms of algebraic numbers, as well as various constructions and tools from Diophantine approximation and automata theory. It should be noted that the almost periodic and disjunctive cases are genuinely distinct since a non-constant infinite word cannot have both properties, which we show in Prop. 2.7.

The hypotheses of Thm. 1.1 (that $\langle u_n \rangle_{n=0}^{\infty}$ be non-degenerate, simple, and with at most three dominant roots) may at first glance appear quite restric-

tive. It is worth noting, however, that if one were to pick LRS "at random", then under any "reasonable" probability distribution, with probability 1 such LRS would have either a single dominant root, or be non-degenerate, simple, and have exactly two dominant roots. In other words, the vast majority of LRS are of one of these two types, and thus fall under the scope of the Elgot-Rabin contraction method or our own Thm. 1.1.[3] In any event, we briefly return in Sec. 4 to the question of whether the hypotheses of Thm. 1.1 could be weakened in any way.

2. Preliminaries

We write $\mathbb{T}$ for the unit circle $\{z \in \mathbb{C} \colon |z| = 1\}$. For non-zero $z \in \mathbb{C}$, $\arg(z)$ denotes the unique $\theta \in (-\pi, \pi]$ such that $z = |z|e^{i\theta}$. We write $\mathbb{R}/(2\pi\mathbb{Z})$ for the quotient of the group $(\mathbb{R}; +)$ by the subgroup $2\pi\mathbb{Z}$; note that $\mathbb{R}/(2\pi\mathbb{Z})$ is isomorphic to the groups $([0, 2\pi); +)$ (where addition operates modulo 2π) and $(\mathbb{T}; \cdot)$. By the sign of $x \in \mathbb{R}$ we mean one of $+, 0, -$.

2.1. Algebraic numbers

A number $\alpha \in \mathbb{C}$ is *algebraic* if $F(\alpha) = 0$ for a non-zero polynomial $F \in \mathbb{Z}[X]$. The *minimal polynomial* of α is the unique monic polynomial of least degree with this property. Algebraic numbers form a field, written $\overline{\mathbb{Q}}$.

2.2. Automata Theory

An *alphabet* Σ is a finite, non-empty set of letters, and Σ^* and Σ^ω denote the sets of finite and infinite words over Σ, respectively. As words can be viewed as sequences, we freely identify the finite word $w_0 w_1 \cdots w_k$ with the finite sequence $\langle w_0, w_1, \ldots, w_k \rangle$ and the infinite word $w_0 w_1 \cdots$ with the infinite sequence $\langle w_0, w_1, \ldots \rangle$, and conversely. A finite word $w' = w'_0 w'_1 \cdots w'_k$ is a *factor* of an (in)finite word w if there is an index n such that $\langle w_n, \ldots, w_{n+k} \rangle = \langle w'_0, \ldots, w'_k \rangle$. We let $|w|$ denote the length of a finite word w.

An infinite word w is *recursive* (or *computable*) if some Turing machine outputs w_j upon reading j. A word $w \in \Sigma^\omega$ is *ultimately periodic* if $w = uv^\omega$

[3] This folklore result follows from the fact that a random polynomial with real coefficients will almost surely have either a single dominant real root, or two dominant complex conjugate roots; and that non-degeneracy and simplicity are null-measure conditions. For a more in-depth discussion of these matters, see the work of Dubickas and Sha [22] and citations therein.

for some $u \in \Sigma^*$ and non-empty $v \in \Sigma^*$. Here, we call the lengths of the shortest possible such u and v the *preperiod* and *period* of w, respectively.

A *deterministic Muller automaton* $\mathcal{A} = (\Sigma, Q, q_{\text{init}}, \delta, \mathcal{F})$ consists of an alphabet Σ, a finite set of states Q, an initial state $q_{\text{init}} \in Q$, a transition function $\delta\colon Q \times \Sigma \to Q$, and an accepting family of sets $\mathcal{F} \subseteq 2^Q$. The *run* of $w \in \Sigma^\omega$ on $\mathcal{A}$ is the sequence of states visited starting in q_{init} and updating the state using the transition function while reading letters of w. Then $\mathcal{A}$ *accepts* w if the set of states visited infinitely often belongs to $\mathcal{F}$. The *acceptance problem* of a recursive word $w \in \Sigma^\omega$, denoted Acc_w, is the question of determining, given a deterministic Muller automaton $\mathcal{A}$ with alphabet Σ, whether $\mathcal{A}$ accepts w. By hard-coding the start of a run, the following is immediate.

Fact 2.1. *Let $w \in \Sigma^\omega$ be recursive and $N_1, N_2 \in \mathbb{N}$. Then $\text{Acc}_{w_{N_1} w_{N_1+1} \cdots}$ and $\text{Acc}_{w_{N_2} w_{N_2+1} \cdots}$ are Turing-equivalent.*

A *deterministic finite transducer* $\mathcal{B} = (\Sigma_{\text{in}}, \Sigma_{\text{out}}, Q, q_{\text{init}}, \delta)$ is given by an input alphabet Σ_{in}, output alphabet Σ_{out}, set of states Q, initial state $q_{\text{init}} \in Q$, and transition function $\delta\colon Q \times \Sigma_{\text{in}} \to Q \times \Sigma_{\text{out}}^*$. The transducer $\mathcal{B}$ starts in state q_{init}. It reads a word $w \in \Sigma_{\text{in}}^* \cup \Sigma_{\text{in}}^\omega$ and upon reading $a \in \Sigma$ whilst in state q, computes $(q', w') = \delta(q, a)$, moves to state q', and concatenates w' to the output string. For simplicity, we often write transducer for deterministic finite transducer. Let $\mathcal{B}(w) \in \Sigma_{\text{out}}^* \cup \Sigma_{\text{out}}^\omega$ denote the output word computed upon reading w. We now recall an important Turing-reducibility property between word acceptance problems.

Lemma 2.2 ([11, Lem. 4.5])**.** *Let $w \in \Sigma^\omega$ and $\mathcal{B}$ be a deterministic finite transducer. Suppose $\mathcal{B}(w)$ is infinite. Then $\text{Acc}_{\mathcal{B}(w)}$ reduces to Acc_w.*

In this paper, we work with certain special classes of words that play a prominent role in contemporary approaches to proving MSO decidability. First, $w \in \Sigma^\omega$ is *almost periodic* if for every $v \in \Sigma^*$ there is a number R such that either

1. v does not appear in $w_R w_{R+1} w_{R+2} \cdots$, or
2. v appears in every factor in w of length R.

Moreover, w is *effectively almost periodic* if R can be effectively computed from v. The following results are due to Semënov [7].

Theorem 2.3. *Let $w \in \Sigma^\omega$ be recursive and effectively almost periodic. Then Acc_w is decidable.*

Theorem 2.4. *Let $w \in \Sigma^\omega$ be effectively almost periodic and $\mathcal{B}$ a transducer with input alphabet Σ. Then $\mathcal{B}(w)$ is either finite (which can be effectively determined), of effectively almost periodic.*

Next, we discuss *toric words*, which are examples of effective almost periodic words. A toric word u over an alphabet Σ is given by $d \geq 1$, a collection $\{U_a \colon a \in \Sigma\}$ of pairwise disjoint open subsets of $\mathbb{T}^d$, and $s, t \in \mathbb{T}^d$. The nth letter of u is determined by the unique U_a containing st^n. The following is a weakened version of [23, Cor. 5.8].

Theorem 2.5. *Let $s, t \in \mathbb{T} \cap \overline{\mathbb{Q}}$ and $U_1, \ldots, U_k \subset \mathbb{T}$ be open and pairwise disjoint. Suppose the boundary points of each U_i lie in $\overline{\mathbb{Q}}$, and for all $n \in \mathbb{N}$, $st^n \in U_i$ for some $1 \leq i \leq k$. The toric word $w \in \{1, \ldots, k\}^\omega$ generated by $s, t, \{U_i \colon 1 \leq i \leq k\}$ is effectively almost periodic.*

Disjunctive words form our second class of words of interest. A word $w \in \Sigma^\omega$ is *disjunctive* if each $v \in \Sigma^*$ appears infinitely often as a factor of w. Other authors occasionally use the terminology *weakly normal* or *rich* instead of disjunctive. Berthé *et al.* established the following [11, Thm. 4.16].

Theorem 2.6. *If $w \in \Sigma^\omega$ is recursive and disjunctive, then Acc_w is decidable.*

Although the acceptance problem is decidable for both effectively almost periodic and disjunctive words, these classes are orthogonal to each other.

Proposition 2.7. *Assume $|\Sigma| > 1$. Then $w \in \Sigma^\omega$ cannot be both almost periodic and disjunctive.*

Proof. Assume, for a contradiction, that w is almost periodic and disjunctive, and let a and b be distinct letters in Σ. Then, by disjuntivity, a appears infinitely often in w and so, by almost periodicity, there is an R such that a appears in each factor of w of length R. Thus, b^R does not appear in w, contradicting disjunctivity. □

2.3. Monadic Second-Order Logic

A *(unary or monadic) predicate* P is a function $P \colon \mathbb{N} \to \{0, 1\}$, which we equivalently interpret as a subset $P \subseteq \mathbb{N}$. For an infinite predicate $P \subseteq \mathbb{N}$, let us write $\langle p_m \rangle_{m=0}^\infty$ to denote the enumeration of P in increasing order, i.e.,

such that p_{m-1} is the mth smallest element of P. The *characteristic word* $w \in \{0,1\}^\omega$ of P is obtained by setting $w_n = P(n)$. Then,

$$w = 0^{p_0}10^{p_1-p_0-1}10^{p_2-p_1-1}10^{p_3-p_2-1}\cdots. \tag{3}$$

A predicate P is *sparse* if for every $N \geq 0$, there is an $M \geq 0$ such that $p_{m+1} - p_m \geq N$ for all $m \geq M$. The predicate is *effectively sparse* if M can be effectively computed from N.

Monadic second-order (MSO) logic is an extension of first-order logic over signature $\{=, <, \in\}$ that allows quantification over subsets of the universe. We consider MSO over expansions of $\mathbb{N}$ by various fixed unary predicates $P \subseteq \mathbb{N}$; in other words (abusing notation), the signature is expanded by a predicate symbol P, with interpretation the given subset $P \subseteq \mathbb{N}$. We refer the reader to Blumensath [24] for a thorough contemporary overview of MSO.

The deep connection between MSO and automata theory was brought to light by Büchi's seminal work; see, e.g., [25, Thms. 5.4 and 5.9].

Theorem 2.8. *The decision problem of the MSO theory of the structure $\langle \mathbb{N}; <, P\rangle$ (i.e., given an MSO formula φ, decide whether it holds in the structure) is Turing-equivalent to Acc_w, where w is the characteristic word of P.*

Elgot and Rabin devised the *contraction method* to establish decidability of the MSO theory of $\langle \mathbb{N}; <, P\rangle$ for various predicates P of arithmetic origin. The following proposition is a variation on their method:

Proposition 2.9. *Let $P \subseteq \mathbb{N}$ be an infinite, recursive, and effectively sparse predicate with enumeration $\langle p_m\rangle_{m=0}^\infty$. If, for each $M \geq 1$, $\mathrm{Acc}_{\langle p_m \bmod M\rangle_{m=0}^\infty}$ is decidable, then the MSO theory of the structure $\langle \mathbb{N}; <, P\rangle$ is decidable.*

Proof. By Thm. 2.8, it is sufficient to be able to decide whether a deterministic Muller automaton $\mathcal{A} = (\{0,1\}, Q, q_{\mathrm{init}}, \delta, \mathcal{F})$ accepts the characteristic word (3) of P. Let us restrict $\mathcal{A}$ to a directed graph G with nodes Q and 0-transitions as arrows.

By construction, every vertex in G has outdegree 1 and so each vertex is in at most one cycle in G. We can therefore compute the least common multiple of the cycle lengths in G (call this number M) and the longest path to a cycle (call this number N). Then, for all states q and numbers $n \geq M + N$ and $d \geq 1$, reading 0^n and 0^{n+dM} lead to journeying through the same set of states and ending up in the same state.

Compute, using effectively sparsity, K such that $p_{m+1} - p_m - 1 \geq M + N$ for all $m \geq K$. We construct a deterministic finite transducer $\mathcal{B}$ that hard-codes the initial segment $w_{\text{init}} := 0^{p_0} 1 0^{p_1 - p_0 - 1} 1 \cdots 0^{p_{K+1} - p_K - 1} 1$. For $m \geq K$, after reading $p_m \bmod M$ and $p_{m+1} \bmod M$, we let $\mathcal{B}$ output $0^{k_m} 1$, where $M + N < k_m \leq 2M + N$ is congruent to $p_{m+1} - p_m - 1$ modulo M. Then, by construction, a state q is visited infinitely often upon reading w if and only if q is visited infinitely often when $\mathcal{A}$ reads $w_{\text{init}} \mathcal{B}(\langle p_m \bmod M \rangle_{m=K+1}^{\infty})$. Thus, using Fact 2.1, Acc_w is decidable when $\text{Acc}_{\mathcal{B}(\langle p_m \bmod M \rangle_{m=K+1}^{\infty})}$ is decidable.

As $\text{Acc}_{\langle p_m \bmod M \rangle_{m=0}^{\infty}}$ is assumed to be decidable, $\text{Acc}_{\langle p_m \bmod M \rangle_{m=K+1}^{\infty}}$ is also decidable (by Fact 2.1) and thus $\text{Acc}_{\mathcal{B}(\langle p_m \bmod M \rangle_{m=K+1}^{\infty})}$ is decidable by Thm. 2.2. Therefore Acc_w is decidable, as required. □

2.4. Linear recurrence sequences

A *linear recurrence sequence* over a ring R (an *R-LRS*) is an R-valued sequence $\boldsymbol{u} = \langle u_n \rangle_{n=0}^{\infty}$ for which there exist fixed coefficients $c_1, \ldots, c_d \in R$ with $c_d \neq 0$ such that, for all $n \in \mathbb{N}$, the terms satisfy the recurrence

$$u_{n+d} = c_1 u_{n+d-1} + \cdots + c_d u_n \,.$$

These $2d$ numbers $c_1, \ldots, c_d, u_0, \ldots, u_{d-1}$ completely specify the entire sequence. Among all such recurrence relations, we henceforth consider only the unique one with minimal order d. In the remainder of the paper, $R = \mathbb{Z}$ whenever R is unspecified. The *characteristic polynomial* of the LRS $\boldsymbol{u}$ is given by $F(X) = X^d - c_1 X^{d-1} - \cdots - c_d \in R[X]$. The *characteristic roots* of $\boldsymbol{u}$ are the roots of its characteristic polynomial, and the *multiplicity* of a characteristic root λ is its multiplicity as a root of $F(X)$. Every $\overline{\mathbb{Q}}$-LRS $\boldsymbol{u}$ with characteristic roots $\lambda_1, \ldots, \lambda_K$ admits a unique *exponential-polynomial* representation given by

$$u_n = \sum_{k=1}^{K} Q_k(n) \lambda_k^n \,,$$

where $Q_k \in \overline{\mathbb{Q}}[X]$ has degree the multiplicity of λ_k minus 1. If $\lambda_k \in \mathbb{R}$ then Q_k has real coefficients, and if $\lambda_{k'} = \overline{\lambda_k}$ then $Q_k = \overline{Q_{k'}}$ (where the complex conjugate of a polynomial is obtained by taking complex conjugates of the coefficients). A $\overline{\mathbb{Q}}$-LRS is *simple* if all its characteristic roots have multiplicity 1, and is *non-degenerate* if every quotient γ of two distinct characteristic roots is not a root of unity, i.e., satisfies $\gamma^k \neq 1$ for all non-zero integers k.

A characteristic root λ is *dominant* if $|\lambda| \geq |\mu|$ for all characteristic roots μ. We define the *dominant part* $\boldsymbol{v}$ of $\boldsymbol{u}$ by writing

$$v_n = \sum_{\lambda_j \text{ dominant}} Q_j(n)\lambda_j^n\,,$$

with the *non-dominant part* $\boldsymbol{r}$ given by $r_n = u_n - v_n$. If $\boldsymbol{u}$ is a $(\overline{\mathbb{Q}} \cap \mathbb{R})$-LRS, then its dominant and non-dominant parts are both $(\overline{\mathbb{Q}} \cap \mathbb{R})$-LRS. One can compute positive real numbers r and R such that $rR^n > |r_n|$ for all $n \in \mathbb{N}$ and $R < |\lambda|$ for λ a dominant characteristic root, see, e.g., [11, Lem. 2.5].

If $M \geq 1$ and $\boldsymbol{u}$ is a $\mathbb{Z}$-LRS, the sequence $\langle u_n \bmod M\rangle_{n=0}^{\infty}$ is ultimately periodic, and one can effectively compute its period and preperiod [11, Lem. 2.6].

Example 2.10. *Let $\boldsymbol{u}$ be an LRS satisfying* (1)*. Then*

$$F(X) = X^3 - 3X^2 + 6X - 8 = \big(X-2\big)\big(X-\lambda\big)\big(X-\overline{\lambda}\big)$$

is the characteristic polynomial of $\boldsymbol{u}$. Its characteristic roots are 2*, λ, and $\overline{\lambda}$, where $\lambda = \frac{1+\sqrt{-15}}{2}$. Such LRS are simple and non-degenerate with exponential-polynomial representation is $u_n = \alpha\lambda^n + \overline{\alpha}\overline{\lambda}^n + \beta 2^n$ for some algebraic numbers α and β. As all characteristic roots are dominant, the dominant part of $\boldsymbol{u}$ equals $\boldsymbol{u}$ itself, and its non-dominant part is the sequence that is constantly* 0*.*

2.5. Number theory

Our first number-theoretical tool is Kronecker's theorem on Diophantine approximation (see, e.g. [26]).

Theorem 2.11. *Suppose $\theta \in \mathbb{R}$ is not a rational multiple of π. Then $\{\theta n \bmod 2\pi \colon n \in \mathbb{N}\}$ is dense in $\mathbb{R}/(2\pi\mathbb{Z})$.*

Our second number-theoretic tool is Baker's theorem on linear forms in logarithms, see, e.g., [27].

Theorem 2.12. *Suppose $\alpha_1, \ldots, \alpha_M \in \overline{\mathbb{Q}} \setminus \{0\}$. Then there is a computable $c > 0$ such that for any $b_1, \ldots, b_M \in \mathbb{Z}$, writing $B = \max\{|b_1|, \ldots, |b_M|, 2\}$ and $\Lambda = \prod_{j=1}^{M} \alpha_j^{b_j} - 1$, if $\Lambda \neq 0$ then $|\Lambda| > B^{-c}$.*

We will also need an off-the-shelf application of Baker's theorem due to Mignotte, Shorey, and Tijdeman [28]:

Theorem 2.13. *Let $\boldsymbol{u}$ be a non-degenerate LRS with at most three dominant roots of magnitude $|\lambda|$. Then there are computable positive constants $C_1, \ldots, C_4$ such that*

1. $|u_n| \geq |\lambda|^n \cdot n^{-C_1 \log(n)}$ *whenever* $n \geq C_2$*; and*
2. $|u_{n_1} - u_{n_2}| \geq |\lambda|^{n_1} \cdot {n_1}^{-C_3 \log(n_1) \log(n_2+2)}$ *whenever* $n_1 > n_2$ *and* $n_1 \geq C_4$.

3. Proof of the Main Result

In this section, we prove Thm. 1.1: the MSO theory of the structure $\langle \mathbb{N}; <, P \rangle$ is decidable for predicates P comprising the set of positive terms of a non-degenerate, simple LRS with at most three dominant characteristic roots. In this proof, we also establish Thms. 1.3 and 1.5. As decidability for LRS with a single dominant root follows from the classical work of Elgot and Rabin, we only need to study LRS with two or three dominant roots.

In Sec. 3.1, we unpack the definition of an LRS satisfying the above hypotheses and reduce the problem to three cases. The first case is trivial, since the predicate is finite. The second, treated in Sec. 3.2, uses almost periodicity and is slightly more involved. The bulk of the paper is devoted to the third case, handled in Sec. 3.3, where we rely on prodisjunctivity.

3.1. Reduction to the almost periodic and prodisjunctive cases

Let $\boldsymbol{u}$ be an LRS satisfying the hypotheses of Thm. 1.1. We first record some elementary observations.

Lemma 3.1. *Assume that $\boldsymbol{u}$ is a simple, non-degenerate LRS with two or three dominant roots. Then we can write*

$$u_n = \alpha\lambda^n + \overline{\alpha}\overline{\lambda}^n + \beta\rho^n + r_n = v_n + r_n$$

for some computable $\alpha \in \overline{\mathbb{Q}} \setminus \{0\}$, $\beta \in \overline{\mathbb{Q}} \cap \mathbb{R}$, and $\lambda \in \overline{\mathbb{Q}}$ such that $\arg(\lambda)$ *is not a rational multiple of π, $|\lambda| > 1$, $\rho = \pm|\lambda|$, and $\boldsymbol{v}$ and $\boldsymbol{r}$ are the dominant and non-dominant parts of $\boldsymbol{u}$, respectively.*

Proof. Let λ and λ' be two dominant roots of $\boldsymbol{u}$. Thus, $|\lambda| = |\lambda'|$. If both roots are real, $\lambda/\lambda' = \pm 1$, contradicting non-degeneracy, and thus, without loss of generality, λ is non-real. Hence, as the exponential-polynomial form of an LRS is closed under complex-conjugation, $\overline{\lambda}$ is also a dominant root.

If $\arg(\lambda)$ is a rational multiple of π, then $\lambda/\overline{\lambda}$ is a root of unity, contradicting non-degeneracy. If $|\lambda| \leq 1$, then as λ is dominant, the product of the

characteristic roots, call it c, is at most 1 but non-zero. However, $\pm c$ also equals the constant coefficient of the integer-valued characteristic polynomial of $\boldsymbol{u}$. Thus $c = \pm 1$ and all characteristic roots are dominant with absolute value 1. Hence, by a result of Kronecker [29], λ is a root of unity, which again contradicts non-degeneracy. Thus, $|\lambda| > 1$. Lastly, as $\boldsymbol{u}$ is simple, the polynomial coefficient of λ in the exponential-polynomial form of $\boldsymbol{u}$ is a non-zero constant, which we can call α.

If $\boldsymbol{u}$ has two dominant roots, set $\rho = |\lambda|$ and $\beta = 0$. Otherwise, let ρ be the third characteristic root of $\boldsymbol{u}$. Then ρ must be real: Otherwise, $\overline{\rho}$ would be an impossible fourth characteristic root. Thus, as ρ and λ are both dominant, $\rho = \pm|\lambda|$ and the polynomial coefficient of ρ has to be real as the exponential-polynomial form is closed under complex conjugation. □

For the remainder of the paper, fix α, β, λ, ρ and $\boldsymbol{r}$ to be as in the lemma above. Further fix $P := \{u_n \colon n \in \mathbb{N}\} \cap \mathbb{N}$.

To decide the MSO theory of $\langle \mathbb{N}; <, P \rangle$, it is crucial to understand when $u_n \geq 0$ and when $u_n \geq u_m$. Luckily, by the results of Mignotte, Shorey, and Tijdeman given in Sec. 2.4, this information can be inferred from the ordering of the non-negative terms of $\boldsymbol{v}$.

Lemma 3.2. *We can effectively compute $N \in \mathbb{N}$ such that for all $n \geq N$ and $m \geq 0$,*

1. *u_n and v_n have the same sign;*
2. *$u_n - u_m$ and $v_n - v_m$ have the same sign.*

Moreover, P is recursive and effectively sparse.

Proof. Compute $0 < R < |\lambda|$ and $r > 0$ such that $rR^n \geq |r_n|$ for all $n \in \mathbb{N}$. Let $C_1, C_2, C_3, C_4 > 0$ be the computable constants from Thm. 2.13.

Suppose the signs of u_n and v_n differ. Then $|u_n| \leq |r_n|$ and therefore $|u_n| \leq rR^n$. Applying Thm. 2.13, either $n < C_2$, or

$$|\lambda|^n n^{-C_1 \log(n)} \leq |u_n| \leq rR^n \,.$$

Since $|\lambda|^n n^{-C_1 \log(n)}$ grows asymptotically faster than rR^n, $n < D_1$ for a computable constant D_1. Taking $N \geq \max\{C_2, D_1\}$ gives item 1. The recursiveness of P follows similarly. Suppose we want to compute all n such that $|u_n| < |C|$ for a given constant C. We first compute D_2 such that for all $n \geq D_2$, $|\lambda|^n n^{-C_1 \log(n)} \geq C$. Then we check all $n < \max\{C_2, D_2\}$.

To prove item 2, assume without loss of generality that $n > m$: if $n = m$, then the conclusion is immediate, and if $m > n$, then $m \geq N$, and we can swap the roles of n, m, i.e., prove that $u_m - u_n$ and $v_m - v_n$ have the same sign. Suppose the signs of $u_n - u_m$ and $v_n - v_m$ are different. Since $u_n - u_m = v_n - v_m + (r_n - r_m)$, it must be the case that

$$|u_n - u_m| \leq |r_n - r_m| \leq rR^n + rR^m < 2rR^n.$$

Applying Thm. 2.13, either $n < C_4$, or

$$|\lambda|^n n^{-C_3 \log(n) \log(n+2)} \leq |u_n - u_m| \leq 2rR^n.$$

This cannot hold for n larger than or equalt to a computable constant D_3, and we can take $N = \max\{D_3, C_4\}$.

Finally, we prove effective sparsity. Suppose, given a constant C, we want to find all $n, m \in \mathbb{N}$, $n > m$ such that $|u_{n_1} - u_{n_2}| < C$. We compute D_4 such that $|\lambda|^n n^{-C_3 \log(n) \log(n+2)} \geq C$, and enumerate all n_1, n_2 such that $n_2 < n_1 < \max\{D_4, C_4\}$. □

In our situation, P is potentially finite. Luckily, in that case, we can also effectively compute P and thus decide the MSO theory of $\langle \mathbb{N}; <, P \rangle$.

Lemma 3.3. *We have that P is finite if and only if $\rho > 0$ and $\beta \leq -2|\alpha|$, and when P is finite, it can be effectively computed. Furthermore, without loss of generality, $|\beta| < 2|\alpha|$ whenever $\rho < 0$.*

Proof. If $\rho > 0$ and $\beta \leq -2|\alpha|$, then $\rho = |\lambda|$ and

$$v_n = \alpha\lambda^n + \overline{\alpha}\overline{\lambda}^n + \beta\rho^n \leq (2|\alpha| - \beta)|\lambda|^n \leq 0\,.$$

Applying Lem. 3.2, we have that for all sufficiently large n (with a computable threshold N), $u_n \geq 0$ if and only if $u_n = v_n = 0$. Using the same lemma, we check whether $0 \in P$. In this way, we can compute P effectively, which is indeed finite.

To prove that P is infinite in any other case, we again use Lem. 3.2. Then, writing $\alpha = |\alpha|e^{i\varphi}$ and $\lambda = |\lambda|e^{i\theta}$ it is sufficient to show that

$$\frac{v_n}{|\lambda|^n} = \frac{1}{|\lambda|^n}(\alpha\lambda^n + \overline{\alpha}\overline{\lambda}^n + \beta\rho^n) = 2|\alpha|\cos(n\theta + \varphi) + \beta\left(\frac{\rho}{|\lambda|}\right)^n$$

is dense in an open interval $\mathcal{I} \subset \mathbb{R}_{>0}$. Using Thm. 2.11, $\cos(2n\theta + \varphi)$ and $\cos((2n+1)\theta + \varphi)$ are dense in $[-1, 1]$. Here, $(\rho/|\lambda|)^n$ is either 1 (when

$\rho > 0$) or $(-1)^n$ (when $\rho < 0$). In the former case, $v_n/|\lambda|^n$ is dense in the interval $[-2|\alpha| + \beta, 2|\alpha| + \beta]$ which contains an open interval $\mathcal{I} \subset \mathbb{R}_{>0}$ as $2|\alpha| > -\beta$. In the second case, one of $v_{2n}/|\lambda|^{2n}$ and $v_{2n+1}/|\lambda|^{2n+1}$ is dense in $I_1 = [-2|\alpha| + |\beta|, 2|\alpha| + |\beta|]$, which again contains the required interval $\mathcal{I}$, while the other subsequence is dense in $I_2 = [-2|\alpha| - |\beta|, 2|\alpha| - |\beta|]$. Thus, P is indeed infinite.

Now suppose $\rho < 0$ and $|\beta| \geq 2|\alpha|$. Note that $\langle u_{2n}\rangle_{n=0}^{\infty}$ and $\langle u_{2n+1}\rangle_{n=0}^{\infty}$ are both LRS with a real dominant root $\rho^2 > 0$. For simplicity, let us also assume $\beta > 0$; the case of $\beta < 0$ is similar. Then $I_2 \cap \mathbb{R}_{>0} = \varnothing$, and v_{2n+1} is never positive. Applying Lem. 3.2, we have that u_{2n+1} takes on finitely many non-negative values, which can can be effectively determined as argued earlier. Hence the decision problems of MSO theories of $\langle \mathbb{N}; <, P\rangle$ and $\langle \mathbb{N}; <, \{u_{2n} \colon n \in \mathbb{N}\}\rangle$ are Turing-equivalent. □

Thanks to this lemma, we assume from now on that P is infinite, and that $|\beta| < 2|\alpha|$ when $\rho < 0$. Let $\boldsymbol{p} := \langle p_m\rangle_{m=0}^{\infty}$ enumerate P in increasing order. By Prop. 2.9, it suffices to prove that $\mathrm{Acc}_{\langle p_m \bmod M\rangle_{m=0}^{\infty}}$ is decidable for all $M \geq 1$, and it remains to understand the sequences $\boldsymbol{p}^{(M)} := \langle p_m \bmod M\rangle_{m=0}^{\infty}$.

Now only two cases of our trichotomy remain, namely the almost periodic case of Thm. 1.3 and the disjunctive case of Thm. 1.5. We can rewrite the hypotheses of these theorems easily in terms of α and β. First, suppose $\rho > 0$ and $\beta > 2|\alpha|$. Then $0 < (\beta - 2|\alpha|)\rho^n \leq v_n \leq (\beta + 2|\alpha|)\rho^n$, and for any $D > 0$ satisfying $|\lambda|^D > \frac{\beta+2|\alpha|}{\beta-2|\alpha|}$, we have that $v_n > v_{m+D}$ whenever $n > m + D$. The remaining cases are $-2|\alpha| < \beta < 2|\alpha|$, as well as $\rho > 0$ and $\beta = 2|\alpha|$. In both of these cases, write $v_{2n} = (2|\alpha|\cos(\theta n + \varphi) + \beta)|\lambda|^{2n}$, and a density argument using Thm. 2.11 shows that $2|\alpha|\cos(\theta n + \varphi) + \beta$ can achieve arbitrarily small positive values. Hence, for any $D \geq 0$, there are $n > m + D$ such that $0 < v_n < v_m$, and the equivalent statement using $\boldsymbol{u}$ follows by Lem. 3.2.

Lemma 3.4. *Theorems 1.3 and 1.5 imply Thm. 1.1, i.e., if $\boldsymbol{u}$ is an LRS satisfying the conditions of Thm. 1.1 and $P := \{u_n \colon n \in \mathbb{N}\} \cap \mathbb{N}$, then the MSO theory of the structure $\langle \mathbb{N}; <, P\rangle$ is decidable.*

Proof. When P is finite, Lem. 3.3 shows that P can be computed effectively, and so the MSO theory of $\langle \mathbb{N}; <, P\rangle$ is decidable. Thus, we can assume that P is infinite, and so by Lem. 3.2, P is recursive and effectively sparse. Then, by Prop. 2.9, we only need to verify that $\mathrm{Acc}_{\boldsymbol{p}^{(M)}}$ is decidable for $M \geq 1$.

By the case analysis above, $\rho > 0$ and $\beta > -2|\alpha|$, or $\rho < 0$ and $|\beta| < 2|\alpha|$. When $\rho > 0$ and $\beta > 2|\alpha|$, the argument above shows that $\boldsymbol{u}$ satisfies the

hypothesis of Thm. 1.3, giving that $\boldsymbol{p}^{(M)}$ is effectively almost periodic for all $M \geq 1$, and thus $\mathrm{Acc}_{\boldsymbol{p}^{(M)}}$ is decidable for $M \geq 1$ by Thm. 2.3. In the remaining cases, the argument above shows that the hypothesis of Thm. 1.5 is satisfied, and hence $\boldsymbol{p}$ is effectively prodisjunctive. Fix $M \geq 1$, and compute S_M and N_M as in the definition of effective prodisjunctivity. Further, compute N' such that $p_{N'} > u_0, \ldots, u_{N_M - 1}$. Then, $\langle p_m \bmod M \rangle_{m=N'}^{\infty} \in S_M^{\omega}$ is disjunctive. Let $\mathcal{A}$ be a deterministic Muller automaton over the alphabet $\{0, \ldots, M-1\}$ and $\mathcal{A}'$ the restriction of $\mathcal{A}$ to the alphabet S_M. By Thm. 2.6, we can decide whether $\mathcal{A}'$ accepts $\langle p_m \bmod M \rangle_{m=N'}^{\infty}$ and so whether $\mathcal{A}$ accepts $\langle p_m \bmod M \rangle_{m=N'}^{\infty}$. Thus, by Fact 2.1, we can decide whether $\mathcal{A}$ accepts $\boldsymbol{p}^{(M)}$, and so $\mathrm{Acc}_{\boldsymbol{p}^{(M)}}$ is decidable. It remains to apply Prop. 2.9. $\square$

3.2. The almost periodic case

We now prove Thm. 1.3. As argued above, the hypothesis is equivalent to $\rho > 0$ and $\beta > 2|\alpha|$.

Proof of Thm.1.3. We construct a toric word w such that for every $M \geq 1$, we can construct a transducer $\mathcal{B}$ such that $\mathcal{B}(w) = \boldsymbol{p}^{(M)}$.

Recall that $v_n = \alpha\lambda^n + \overline{\alpha}\overline{\lambda}^n + b\rho^n$, where $\rho > 0$ and $b > 2|\alpha|$. As we are only interested in the ordering of the v_n relative to 0 and each other, divide by $2|\alpha|$ to obtain $v_n = (\cos(n\theta + \varphi) + b)\rho^n$ for $b > 1$, $\alpha = e^{i\varphi}$, and $\lambda = \rho e^{i\theta}$.

Let $\mathbb{T}$ denote the unit circle $\{z \in \mathbb{C} \colon |z| = 1\}$ and $U \subset \mathbb{T}$ be the set such that $x \in U$ if and only if $\cos(x) + b > (\cos(x - \theta n) + b)\rho^{-n}$ for all $n \geq 1$. When $\cos(x)$ is close to 1, $x \in U$ as $b, \rho > 1$. Thus, U is non-empty. We claim that U is an open interval whose boundary consists of computable algebraic numbers. Compute D such that $\rho^D > \frac{b+1}{b-1}$. Then, for all $n \geq D$,

$$\cos(x) + b \geq b - 1 > (1+b)\rho^{-D} \geq (1+b)\rho^{-n} \geq (\cos(x - \theta n) + b)\rho^{-n}\,.$$

Thus, we only need to consider $1 \leq n < D$. Then, if $\cos(x) + b = (\cos(x - \varphi n) + b)\rho^{-n}$, the quadratic equation

$$(\lambda^{-n} - 1)e^{2ix} + b(\rho^{-n} - 1)e^{ix} + \overline{\lambda}^{-n} - 1 = 0\,,$$

has at most two solutions e^{ix} in $\mathbb{T}$, which are also algebraic numbers. Thus, U is the intersection of at most $D - 1$ intervals with algebraic boundaries. This proves our claim.

Next, compute a number E such that for all $x \in \mathbb{T}$, at least one of $e^{ix}, e^{i(x+\theta)}, \ldots, e^{i(x+(E-1)\theta)}$ is in U, as follows. Let $E' > 1/(2\pi|U|)$. Then there

exist $0 \le c_1 < c_2 \le E'$ such that $e^{i\theta(c_2-c_1)}$ has arc length shorter than $|U|$. By non-degeneracy, $e^{i\theta(c_2-c_1)} \neq 0$. Then one of $e^{i\theta(c_2-c_1)}, \dots, e^{i\theta(c_2-c_1)\lfloor(2\pi(c_2-c_1)\theta)^{-1}\rfloor}$ is in U and so set $E = (c_2 - c_1)\lfloor(2\pi(c_2 - c_1)\theta)^{-1}\rfloor$.

Now consider all possible sequences $\boldsymbol{a} = \langle 0, a_1, \dots, a_e\rangle$ of distinct numbers $1 \le a_i \le E$. Let $U_{\boldsymbol{a}} \subset U$ be such that $x \in U_{\boldsymbol{a}}$ if and only if

- $\cos(x) + b < (\cos(x + a_1\theta) + b)\rho^{a_1} < \cdots < (\cos(x + a_e\theta) + b)\rho^{a_e}$;
- $x + a_i\theta \notin U$ for $1 \le i \le e$; and
- for all $k \in \mathbb{N}$, $\cos(x) + b < (\cos(x + k\theta) + b)\rho^k < (\cos(x + a_e\theta) + b)\rho^{a_e}$ implies that $k \in \{a_1, \dots, a_{e-1}\}$.

Then, by the same reasoning as for U, each $U_{\boldsymbol{a}}$ is open with a boundary consisting of computable algebraic numbers. Translating the cosines into expressions in v_n and using Lem. 3.2, we can compute N such that if $n \ge N$, $e^{i(n\varphi+\theta)} \in U_{\boldsymbol{a}}$, and $p_m = u_n$, then $(p_m, \dots, p_{m+e}) = (u_n, u_{n+a_1}, \dots, u_{n+a_e})$.

Let $U' = \mathbb{T} \setminus \overline{U}$ be the open complement of U. Further let w be the toric word generated by the rotation $e^{i(n\theta+\varphi)}$ and the open sets U' and all $U_{\boldsymbol{a}}$, i.e., $w_n = \boldsymbol{a}$ if $e^{i(n\theta+\varphi)} \in U_{\boldsymbol{a}}$ and $w_n = \varepsilon$ otherwise. Then, for $n \ge N$, we have that $w_n = \varepsilon$ implies that $e^{i(n\theta+\varphi)} \in U'$. Thus, by Thm. 2.5, w is effectively almost periodic.

Fix $M \ge 1$ and compute the period T and preperiod T' of $\langle u_n \bmod M\rangle_{n=0}^{\infty}$ and update N such that $T' < N$. Then we build a transducer $\mathcal{B}$ such that $\mathcal{B}(w) = \boldsymbol{p}^{(M)}$ as follows: First, hardcode the initial terms $p_0 \bmod M, \dots, p_{N'-1} \bmod M$ such that $p_{N'} = u_N$. Then when reading w_n, two possibilities can occur: First, if $w_n = \varepsilon$ or $n < N$, we continue without changing the state of the transducer or outputting a word. Second, when reading $w_n = \langle 0, a_1, \dots, a_e\rangle$, output $\langle u_n \bmod M, u_{n+a_1} \bmod M, \dots, u_{n+a_e} \bmod M\rangle$, which we can compute as we only need to know $n \bmod T$ which a transducer can easily track. Then, by construction, $\mathcal{B}(w) = \boldsymbol{p}^{(M)}$, and so as w is effectively almost periodic, $\boldsymbol{p}^{(M)}$ is effectively almost periodic by Thm. 2.4. $\square$

3.3. The disjunctive case

Now consider Thm. 1.5. It asserts that, for any $M \ge 1$, the sequence $\langle p_m \bmod M\rangle_{m=N_M}^{\infty} \in {S_M}^{\omega}$ is disjunctive for some computable number N_M and a computable set $S_M \subset \{0, \dots, M-1\}$. Let us unpack this definition. By definition, $\langle p_m \bmod M\rangle_{m=N_M}^{\infty} \in {S_M}^{\omega}$ is disjunctive if for any $\ell \ge 1$, every pattern $\langle s_1, \dots, s_\ell\rangle \in {S_M}^{\ell}$ appears infinitely often in $\langle p_m \bmod M\rangle_{m=N_M}^{\infty}$.

That is, for all $\ell, N \in \mathbb{N}$ with $N \geq N_M$ and $s_1, \ldots, s_\ell \in S_M$, there are $n_1, \ldots, n_\ell \in \mathbb{N}$ such that

1. $n_1, \ldots, n_\ell \geq N$;
2. $u_{n_i} \equiv s_i \pmod{M}$ for all $1 \leq i \leq \ell$;
3. $0 \leq u_{n_1} < \cdots < u_{n_\ell}$;
4. if $m \geq 0$ and $u_{n_1} \leq u_m \leq u_{n_\ell}$, then $u_m \in \{u_{n_1}, \ldots, u_{n_\ell}\}$.

We prefer to work with $\boldsymbol{v}$ rather than $\boldsymbol{u}$. Moreover, we prefer to reason about the difference of $b_\cdot = n_i - n_1$ instead of many different numbers n_i. In the following lemma, we show that we can actually effect these changes.

Lemma 3.5. *Suppose the following holds. Let $T \geq 1$ be even, $\ell \geq 1$, and $t_1, \ldots, t_\ell \in \{0, \ldots, T-1\}$. Then there exist $b_2, \ldots, b_\ell \in \mathbb{N}$ such that, for infinitely many $n \in \mathbb{N}$,*

1. $n \equiv t_1 \pmod{T}$*;*
2. $b_j \equiv t_j \pmod{T}$ *for all* $2 \leq j \leq \ell$*;*
3. $0 < v_n < v_{n+b_2} < \cdots < v_{n+b_\ell}$*;*
4. $m - n \in \{b_2, \ldots, b_{\ell-1}\}$ *whenever* $v_n < v_m < v_{n+b_\ell}$.

Then $\boldsymbol{p}$ is effectively prodisjunctive.

Proof. Fix $M \geq 1$. As described in Sec. 2.4, we can compute the period T and preperiod T' of $\langle u_n \bmod M \rangle_{n=0}^{\infty}$. If T is odd, double T, which is again a period of $\langle u_n \bmod M \rangle_{n=0}^{\infty}$. Then $S_M = \{u_n \bmod M \colon T' < n \leq T' + T\}$. Applying Lem. 3.2 we compute N' such that for all $m_1, m_2 \geq N'$, we have that $u_{m_1} \bmod M \in S_M$ and $u_{m_1} - u_{m_2}$ and $v_{m_1} - u_{m_2}$ have the same sign.

Consider a finite pattern $\langle s_1, \ldots, s_\ell \rangle \in S_m^\ell$, $\ell \geq 1$. We will argue that it appears infinitely often in $\boldsymbol{p}$. For $1 \leq j \leq \ell$, let $r_j \in \{0, \ldots, T-1\}$ be such that for all $n \geq T'$ with $n \equiv r_j \pmod{T}$ we have that $u_n \equiv s_j \pmod{M}$. Define $t_1 = r_1$ and $t_j \equiv r_j - r_1 \pmod{T}$, $t_j \in \{0, \ldots, T-1\}$ for $2 \leq j \leq \ell$. Let $b_2, \ldots, b_\ell \in \mathbb{N}$ satisfy the hypothesis above. Then there exist infinitely many n such that $n \geq \max\{N', T'\}$, $n \equiv t_1 \pmod{T}$, $0 < u_n < u_{n+b_2} < \cdots < u_{n+b_\ell}$, and $m - n \in \{b_2, \ldots, b_{\ell-1}\}$ whenever $u_n < u_m < u_{n+b_\ell}$. Thus, if $p_m = u_{n_1}$, then for $2 \leq j \leq \ell$, we have that $p_{m+j-1} = u_{n+b_j} = u_{k_j}$ for some $k_j \geq T'$ satisfying $k_j \equiv r_j \bmod T$. It follows that $p_{m+j-1} \equiv s_j \pmod{M}$ for all $2 \leq j \leq \ell$. $\square$

In the remainder of this section, we will prove Thm. 1.5 by showing that the hypothesis of Lem. 3.5 is satisfied. As $\alpha \neq 0$ and is only concerned

with inequalities $v_{n_1} < v_{n_2}$ and $v_{n_1} > 0$ for $n_1, n_2 \in \mathbb{N}$, we can scale v_n by $1/|2\alpha|$. That is, we henceforth assume that $|\alpha| = 1/2$. Thus, $|\beta| < 1$ or $\rho = \beta = 1$. Write $\alpha = \frac{1}{2}e^{i\varphi}$, $\lambda = |\lambda|e^{i\theta}$, and $r = \rho/|\lambda| \in \{1, -1\}$. Then $v_n = (\cos(\theta n + \varphi) + \beta r^n)|\lambda|^n$.

Now let us introduce our main technical tool to analyse sequences $\boldsymbol{v}$.

Definition 3.6. *For integers $d \neq 0$, $r = \pm 1$, and real numbers $-1 < b \leq 1$ and $0 \leq \gamma < \delta$ such that $(b, r) \neq (1, -1)$, define $\mathcal{J}_d(\gamma, \delta) \subset \mathbb{R}/(2\pi\mathbb{Z})$ as*

$$\mathcal{J}_d(\gamma, \delta) = \left\{ x \in \mathbb{R}/(2\pi\mathbb{Z}) : \gamma(\cos(x)+b) < (\cos(x+d\theta)+br^d)|\lambda|^d < \delta(\cos(x)+b) \right\}.$$

The utility of this definition is as follows. Let $d > 0$, r be as above. For one of $b = \pm\beta$ (how the value of b is chosen is discussed in the proof of Lem. 3.9), we have that $\theta n + \varphi \in \mathcal{J}_d(\gamma, \delta)$ if and only if $\gamma < v_{n+d}/v_n < \delta$. That is, we can control the ratio of v_n with other v_{n+d} by controlling the intervals $\mathcal{J}_d(\delta, \gamma)$ into which $\theta n + \varphi$ falls. We note that when $\boldsymbol{u}$ has exactly two dominant roots (which is the case in [16]), then $b = 0$ and the technical results about the intervals $\mathcal{J}_d$ that we need to prove are substantially simpler.

Next, we quantify the size of these intervals. For $x \in \mathbb{R}$, let $|x|_{2\pi}$ denote the distance of x to the nearest integer multiple of 2π.

Lemma 3.7. *Let $|b| < 1$. Then there are a finite set $\Xi \subset \mathbb{T}$ and constants $C_6, \ldots, C_9 > 0$ with $C_8 > 1$ that satisfy the following.*

1. *For all $0 \leq \gamma < \delta < C_8$ and $d \geq C_9$, the set $\mathcal{J}_d(\gamma, \delta)$ consists of two intervals $\mathcal{I}$, each satisfying $|\mathcal{I}| < C_6 \frac{\delta - \gamma}{|\lambda|^d}$.*
2. *For any $y \in \mathcal{J}_d(\gamma, \delta)$ and $\chi_\pm = -b \pm \sqrt{b^2 - 1}$, $|y + d\theta - \arg(\chi_\pm)| < C_7|\lambda|^{-d}$.*
3. *For any $\varepsilon > 0$, there is a constant $C_5 > 0$ such that $C_5 \frac{\delta - \gamma}{|\lambda|^d} < |\mathcal{I}|$ provided that $\min_{\xi \in \Xi} |e^{id\theta} - \xi|_{2\pi} > \varepsilon$.*

Proof. Identify $\mathbb{R}/(2\pi\mathbb{Z})$ with $(-\pi, \pi]$. Let $0 \leq \eta \leq C_8$, $f_d(x) = (\cos(x+d\theta) + br^d)|\lambda|^d$, and $g(x) = \eta(\cos(x) + b)$. As $2\cos(x) = e^{ix} + e^{-ix}$ and $e^{id\theta}|\lambda|^d = \lambda^d$,

$$(\cos(x + d\theta) + br^d)|\lambda|^d = \eta(\cos(x) + b) \iff e^{i2x} + 2b\frac{\rho^d - \eta}{\lambda^d - \eta}e^{ix} + \frac{\overline{\lambda}^d - \eta}{\lambda^d - \eta} = 0\,. \tag{4}$$

As the right-hand side is a quadratic equation in e^{ix}, $f_d(x) = g(x)$ has at most two solutions $x \in (-\pi, \pi]$. The last claim now follows from the earlier

claims by taking $\gamma = 0$ and $\delta = C_9$, and noting that $\chi_\pm$ are exactly the two roots of (4) with $\eta = \gamma = 0$. We claim that for small enough $C_8 > 1$, large enough C_9, and $d \geq C_9$, there are exactly two solutions in $(-\pi, \pi]$.

The range of g equals $[-|\eta(1+b)|, |\eta(1+b)|]$, which, for large enough d, is strictly contained in the range of f_d. Then choose C_9 large enough because $d \geq C_9$. Thus, for such d, there are y_-, y_+ such that $f_d(y_-) < g(y_-)$ and $f(y_+) > g(y_+)$, and so using periodicity, continuity, and the mid-value theorem, there are at least, and thus exactly, two x such that $f_d(x) = g(x)$.

We now study the size of $\mathcal{J}_d(\gamma, \delta)$. By the previous argument, $\mathcal{J}_d(\gamma, \delta)$ consists of exactly two intervals $\mathcal{I}$ and $\mathcal{I}'$, whose endpoints are at solutions x_γ, x_δ and x'_γ, x'_δ of (4) for $\eta = \gamma, \delta$, respectively. Thus, without loss of generality, we need to estimate $|\mathcal{I}| = |x_\gamma - x_\delta|_{2\pi}$.

Using the triangle inequality on the unit circle, we get

$$|e^{ix_\gamma} - e^{ix_\delta}| \leq |x_\gamma - x_\delta|_{2\pi} \leq \frac{\pi}{2}|e^{ix_\gamma} - e^{ix_\delta}| \tag{5}$$

and the same holds for x'_γ and x'_δ. Thus, it is sufficient to study $|e^{ix_\gamma} - e^{ix_\delta}|$.

As e^{ix_γ} and e^{ix_δ} approach the same root $\chi_\pm e^{-id\theta}$ of the quadratic equation (4), $|e^{ix_\gamma} - e^{ix_\delta}|$ tends to 0. Similarly, $|e^{ix'_\gamma} - e^{ix'_\delta}|$ tends to 0. These two possibilities for $\chi_\pm$ are $(-br^d \pm \sqrt{b^2-1})e^{id\theta}$ whose difference is thus $2\sqrt{b^2-1}$. Thus, as $b < 1$, there is a constant c_1 such that for large enough d,

$$c_1 < |e^{ix_\gamma} - e^{ix'_\delta}|, |e^{ix'_\gamma} - e^{ix_\delta}| \leq 2\,. \tag{6}$$

Next, through straightforward manipulations (as well as using a symbolic calculator), we verify that

$$\begin{aligned} &|e^{ix_\gamma} - e^{ix_\delta}||e^{ix'_\gamma} - e^{ix'_\delta}||e^{ix_\delta} - e^{ix'_\gamma}||e^{ix'_\delta} - e^{ix_\gamma}| \\ &\qquad = (\delta - \gamma)^2 \frac{|e^{2id\theta} + (2-4b^2)e^{id\theta} + 1||e^{2id\theta} - 1|^2|\lambda|^{2d}}{|\lambda^d - \gamma|^2|\lambda^d - \delta|^2}\,. \end{aligned}$$

Let us add the roots of the polynomials in $e^{i\theta}$ in this numerator to the set Ξ. Moreover, we estimate $|\lambda^d - \eta|$ for $\eta \in \{\gamma, \delta\}$ and sufficiently large d as $c|\lambda|^d < |\lambda^d - \eta| < c'|\lambda|^d$ for some constants $c, c' > 0$. Thus, for some constants $c_2, c_3 > 0$,

$$c_2 \frac{(\delta-\gamma)^2}{|\lambda|^{2d}} < |e^{ix_\gamma} - e^{ix_\delta}||e^{ix'_\gamma} - e^{ix'_\delta}||e^{ix_\delta} - e^{ix'_\gamma}||e^{ix'_\delta} - e^{ix_\gamma}| < c_3 \frac{(\delta-\gamma)^2}{|\lambda|^{2d}}\,. \tag{7}$$

Next, assume $b \neq 0$. Then,

$$\begin{aligned}|e^{ix_\gamma} - e^{ix_\delta}| + |e^{ix'_\gamma} - e^{ix'_\delta}| &\geq |e^{ix_\delta} + e^{ix'_\delta} - e^{ix_\gamma} - e^{ix'_\gamma}| \\ &= 2|b|(\delta - \gamma)\frac{|e^{id\theta} - r^d||\lambda|^d}{|\lambda^d - \gamma||\lambda^d - \delta|}\,.\end{aligned}$$

Then, again, we add $r^d = \pm 1$ to Ξ, and bound $|\lambda^d - \eta|^{-1} < c|\lambda|^{-d}$ for $\eta \in \{\gamma, \delta\}$ for a constant $c > 0$ for large enough d. Here, as $b \neq 0$, there is a constant $c_4 > 0$ such that

$$\max\left\{|e^{ix_\gamma} - e^{ix_\delta}|, |e^{ix'_\gamma} - e^{ix'_\delta}|\right\} \geq \frac{1}{2}\left(|e^{ix_\gamma} - e^{ix_\delta}| + |e^{ix'_\gamma} - e^{ix'_\delta}|\right) > c_4(\delta - \gamma)|\lambda|^{-d}\,. \tag{8}$$

Hence, when $b \neq 0$, combining (6), (7), and (8), we get lower and upper bounds on $|e^{ix_\gamma} - e^{ix_\delta}|$. Substituting these into (5), we indeed recover the first part of the lemma. If $b = 0$, then the quadratic equation (4) has no linear term, so we have that $|e^{ix_\delta} - e^{ix_\gamma}| = |e^{ix'_\delta} - e^{ix'_\gamma}|$. Thus, this combined with (6), (7), and (5), again lets us retrieve the lemma. □

The version of the lemma above with $b = 1$ has slightly different bounds: The sizes decrease "only" by $|\lambda|^{d/2}$ instead of $|\lambda|^d$.

Lemma 3.8. *Let $b = 1$. Then there are a finite set $\Xi \subset \mathbb{T}$ and constants $C_6, \ldots, C_9 > 0$ with $C_8 > 1$ that satisfy the following.*

1. *For all $0 \leq \gamma < \delta < C_8$ and $d \geq C_9$, the set $\mathcal{J}_d(\gamma, \delta)$ consists of two intervals $\mathcal{I}$ each satisfying $|\mathcal{I}| < C_6 \frac{\delta - \gamma}{|\lambda|^{d/2}}$.*
2. *Setting $\chi_\pm = -1$, we have $|y + d\theta - \arg(\chi_\pm)|_{2\pi} < C_7|\lambda|^{-d/2}$ for any $y \in \mathcal{J}_d(\gamma, \delta)$.*
3. *For every $\varepsilon > 0$, there is constant $C_5 > 0$ such that $C_5 \frac{\delta - \gamma}{|\lambda|^{d/2}} < |\mathcal{I}|$ when $\min_{\xi \in \Xi} |e^{i\theta d} - \xi|_{2\pi} > \varepsilon$.*

Proof. The proof is similar to Lem. 3.7. Only, we have to be more careful when deducing that there are two solutions x_η and x'_η to (4). Moreover, there is no formula like (6), and (8) is not strong enough.

As $b = 1$, the definition of the intervals $\mathcal{J}_d$ give that $\rho > 0$ and thus $r = \rho/|\lambda| = 1$. Then let $0 \leq \eta \leq C_8$, $f_d(x) = (\cos(x + d\theta) + 1)|\lambda|^d$, and $g(x) = \eta(\cos(x) + 1)$. We again retrieve (4) and count the number of solutions $x \in (-\pi, \pi]$ of $f_d(x) = g(x)$. Again, by the quadratic equation (4), there are at most two such x. When $\eta = 0$, we need to count the number of solutions

$x \in (-\pi, \pi]$ such that $\cos(x + d\theta) + 1 = 0$, which famously has a unique solution. We will show that when $C_8 > 1$ is small enough, $0 < \eta < C_8$, C_9 is large enough, and $d \geq C_9$, there are exactly two solutions.

The range $[0, 2]$ of g is contained in $[0, 2|\lambda|^d]$, which is the range of f_d. Hence, there is a y_+ such that $f_d(y_+) > g(y_+)$. If $f_d(x) \geq g(x)$ for all $x \in \mathbb{R}$, then $g(x) = 0$ when $f_d(x) = 0$. These occur when $\cos(x + d\theta) = -1$ and $\cos(x) = -1$, respectively, and can only occur simultaneously if θ is not a rational multiple of π, which Lem. 3.1 rules out. Thus, there is a y_- such that $f_d(y_-) < g(y_-)$. Hence, $f_d(x) = g(x)$ for exactly two $x \in (-\pi, \pi]$.

We now tackle the size of the two intervals $\mathcal{I}$ and $\mathcal{I}'$ that compose $\mathcal{J}_d(\gamma, \delta)$. Their endpoints are at solutions x_γ, x_δ and x'_γ, x'_δ of (4) for $\eta = \gamma$ and $\eta = \delta$, respectively. Now it is much easier to write $|\mathcal{I}| = |x_\delta - x_\gamma|_{2\pi} = |x_\delta - x_0|_{2\pi} - |x_\gamma - x_0|_{2\pi}$. Thus, we can freely take $\gamma = 0$. Then (4) has a single solution for $\eta = \gamma = 0$, we have that $x_\gamma = x'_\gamma = -e^{-id\theta}$. Using (7), we now obtain constants $c_1, c_2 > 0$ such that

$$c_1 \frac{\delta}{|\lambda|^d} < |e^{ix_\gamma} - e^{ix_\delta}||e^{ix_\gamma} - e^{ix'_\delta}| < c_2 \frac{\delta}{|\lambda|^d}\,, \tag{9}$$

when we add the appropriate points to Ξ. To replace (6), we compute that

$$|e^{ix_\delta} - e^{ix'_\delta}|^2 = \left|(e^{ix_\delta} + e^{ix'_\delta})^2 - 4e^{ix_\delta}e^{ix'_\delta}\right| = 4\delta \frac{|e^{id\theta} - 1|^2|\lambda|^d|}{|\lambda^d - \delta|^2}\,,$$

suggesting we add 1 to Ξ. Then there is a constant $c_3 > 0$ such that

$$\max\left\{|e^{ix_\delta} - e^{ix_\gamma}|, |e^{ix'_\delta} - e^{ix_\gamma}|\right\} \geq \frac{1}{2}|e^{ix_\delta} - e^{ix'_\delta}| > \frac{c_3}{|\lambda|^{d/2}}\,. \tag{10}$$

If we now combine (9) and (10), we can lower and upper bound the size of $|e^{ix_\delta} - e^{ix_\gamma}|$ and so (5) allows us to bound $|\mathcal{I}|$. □

These lemmas show that the intervals $\mathcal{J}_d$ shrink exponentially fast in d and are exponentially close to at most two specific, rotating points $\chi_\pm e^{-id\theta}$. Therefore, we will refer to $\chi_\pm e^{-id\theta}$ as the *anchor points* of the intervals $\mathcal{J}_d$.

Now let us sketch how we apply these intervals $\mathcal{J}_d$ to prove that the conditions of Lem. 3.5 can be satisfied. In particular, recall that $v_n = (\cos(n\theta + \varphi) + \beta r^n)|\lambda|^n$, and condition 3 of Lem. 3.5: $0 < v_n < v_{n+b_2} < \cdots < v_{n+b_\ell}$. Note that as T is even and $n \equiv t_1 \pmod T$, $r^n = r^{t_1}$, and thus we let $b = r^{t_1}\beta$ Setting $x = n\theta + \varphi$, condition 3 of Lem. 3.5 is equivalent to

$$0 < \cos(x) + b < (\cos(x + b_2\theta) + br^{b_2})|\lambda|^{b_2} < \cdots < (\cos(x + b_\ell\theta) + br^{b_\ell})|\lambda|^{b_\ell}\,.$$

We can strengthen this equation above by adding in some extra terms:

$$\begin{aligned} 0 < \cos(x) + b &< (\cos(x + b_2\theta) + br^{b_2})|\lambda|^{b_2} < \delta_2(\cos(x) + b) \\ &< (\cos(x + b_3\theta) + br^{b_3})|\lambda|^{b_3} < \delta_3(\cos(x) + b) \\ &\quad\vdots \\ &< (\cos(x + b_\ell\theta) + br^{b_\ell})|\lambda|^{b_\ell} < \delta_\ell(\cos(x) + b) \end{aligned} \tag{11}$$

where $\delta_2, \ldots, \delta_\ell$ can be chosen to be any numbers satisfying $1 < \delta_2 < \cdots < \delta_\ell$. We can rewrite this system in terms of the sets $\mathcal{J}_d$ as follows:

$$x \in \mathcal{J}_d(1, \delta_2), \mathcal{J}_d(\delta_2, \delta_3), \ldots, \mathcal{J}_d(\delta_{\ell-1}, \delta_\ell) \;\text{ and }\; \cos(x) + b > 0\,. \tag{12}$$

Similarly, condition 4 of Lem. 3.5 is implied by

$$\forall d \in \mathbb{Z}\colon x \in \mathcal{J}_d(1, \delta_\ell) \Longrightarrow d \in \{0, b_2, \ldots, b_\ell\}. \tag{13}$$

Although we would like to have an interval $\mathcal{I} \subset \mathbb{R}/(2\pi\mathbb{Z})$ where (12) and (13) both hold, this is impossible. Hence, we have to settle for an interval $\mathcal{I}$ where (12) holds and where (13) holds "often". Then, later we find an $x = n\theta + \varphi$ such that (13) is also satisfied. The following lemma is this first step.

Lemma 3.9. *Let $T \geq 1$ be even, $\ell \geq 1$, and $t_2, \ldots, t_\ell \in \{0, \ldots, T-1\}$. Then there are an interval $\mathcal{I} \subset \mathbb{R}/(2\pi\mathbb{Z})$, integers $D, b_2, \ldots, b_\ell \geq 1$, and real numbers $1 < \delta_2 < \cdots < \delta_\ell$ such that*

(a) $b_j \equiv t_j \pmod{T}$ for all $2 \leq j \leq \ell$;
(b) (12) holds for all $x \in \mathcal{I}$;
(c) $\mathcal{I} \cap \mathcal{J}_d(1, \delta_\ell) = \emptyset$ for all integers $d < D$ not in the set $\{b_2, \ldots, b_\ell\}$;
(d) $\sum_{d=D}^{\infty} |\mathcal{J}_d(1, \delta_\ell)| < |\mathcal{I}|$.

When proving that the hypothesis of Lem. 3.5 holds, we will use this lemma as follows. Let $T \geq 1$ be even and $t_1, \ldots, t_\ell \in \{0, \ldots, T-1\}$. Iteratively applying Lem. 3.9, we will construct $1 < \delta_2 < \cdots < \delta_\ell$ and a stack of intervals

$$\mathbb{R}/(2\pi\mathbb{Z}) \supset \mathcal{I}_1 \supset \mathcal{I}_2 := \mathcal{J}_{b_2}(1, \delta_2) \supset \mathcal{I}_3 := \mathcal{J}_{b_3}(\delta_2, \delta_3) \supset \cdots \supset \mathcal{J}_{b_\ell}(\delta_{\ell-1}, \delta_\ell)$$

with $\mathcal{I} = \mathcal{I}_\ell$ as well as $0 < D_1 < \cdots < D_\ell$ with $D = D_\ell$ such that (i) $0 < b_2 < \cdots < b_\ell$, (ii) $D_j > b_j$ for all j, and (iii) the conditions (a-c) are satisfied. Now consider the set of all $n \in \mathbb{N}$ such that $n \equiv t_1 \pmod{T}$ and

$\theta n + \varphi \in \mathcal{I}_\ell$; by Kronecker's theorem, this set is infinite. Using (c) and (d) we will argue that, in fact, for infinitely many such n, we have that $\theta n + \varphi \notin \mathcal{J}_d(1, \delta_\ell)$ for all $d \geq D$. Then, by (b), $v_n > 0$. Moreover, as discussed above $v_{n+b_j}/v_n \in (\gamma_j, \delta_j)$ for all j, and $v_{n+d}/v_n \notin (1, \delta_j)$ for all $d \notin \{b_2, \ldots, b_\ell\}$. It follows that $b_2, \ldots, b_\ell$ satisfy items (1-4) in the statement of Lem. 3.5 for infinitely many $n \in \mathbb{N}$.

We next give a few lemmas necessary for implementing the proof of Lem. 3.5 described above.

Lemma 3.10. *Let $T \geq 1$ and C_8 be the constant from Lems. 3.7 and 3.8. There is a constant $C_{10} > 0$ such that for every $t \in \{0, \ldots, T-1\}$ and small enough interval $\mathcal{I} \subset \mathbb{R}/(2\pi\mathbb{Z})$, there are $|\mathcal{I}|^{-C_{10}} \leq b, n \leq 2|\mathcal{I}|^{-C_{10}}$ such that $n\theta + \varphi \in \mathcal{I}$, one interval of $\mathcal{J}_b(0, C_8)$ is contained in $\mathcal{I}$, and $b, n \equiv t \pmod{T}$.*

Proof. Set $c_1 = \lceil 2\pi|\mathcal{I}|^{-1} \rceil$. By the pigeonhole principle and Thm. 2.11, there are distinct $d_1, d_2 \in \mathbb{N}$ such that $0 \leq d_1, d_2 \leq c_1$ and $|d_1 T\theta - d_2 T\theta|_{2\pi} < |\mathcal{I}|$ and $|d_1 - d_2| < c_1$. Thus, $|(d_1 - d_2)T\theta|_{2\pi} < |\mathcal{I}|$. Note that $e^{i\theta} = \lambda/|\lambda|$ is algebraic. Then, as non-degeneracy prevents $(e^{i\theta T})^{d_1 - d_2} = 1$, Baker's theorem (Thm. 2.12) implies there is a constant c_2 such that

$$\left|(d_1 - d_2)T\theta\right|_{2\pi} > \left|(e^{i\theta T})^{d_1 - d_2} - 1\right| > |d_1 - d_2|^{-c_2} > c_1^{-c_2}\,.$$

Thus, for all $N \in \mathbb{Z}$ and $x \in \mathbb{R}/(2\pi\mathbb{Z})$, there is an $N \leq n' < 2\pi c_1^{c_2} + N$ such that $x + n'(d_1 - d_2)T\theta \in \mathcal{I}$. Setting $n = n'(d_1 - d_2)T + t$ and $x = \theta t + \varphi$, we have that $n\theta + \varphi \in \mathcal{I}$ and $n \equiv t \pmod{T}$ as required. Taking C_{10} slightly bigger than c_2, we have that $2\pi T c_1^{c_2+1} < |\mathcal{I}|^{-C_{10}}$ for small enough $\mathcal{I}$. Then the result for n follows by taking $N = |\mathcal{I}|^{-C_{10}}$.

Let $\mathcal{I}'$ be the middle half of $\mathcal{I}$, χ one of the points $\chi_\pm$, and $\mathcal{J}$ the interval of $\mathcal{J}_b(0, C_8)$ near χ. As before, there is a $b \equiv t \pmod{T}$ such that $|\mathcal{I}'|^{-C_{10}} \leq b \leq 2|\mathcal{I}'|^{-C_{10}}$ and $\chi_\pm e^{-id\theta} \in \mathcal{I}'$. If $\mathcal{J} \not\subseteq \mathcal{I}$, then $\mathcal{J}$ has points inside $\mathcal{I}'$ and outside $\mathcal{I}$. So, $|\mathcal{J}| \geq \frac{1}{4}|\mathcal{I}|$, and Lem. 3.7 implies that

$$\frac{1}{4}|\mathcal{I}| \leq |\mathcal{J}| \leq C_7|\lambda|^{-b} \leq C_7|\lambda|^{-|\mathcal{I}'|^{-C_{10}}} = C_7|\lambda|^{-(|\mathcal{I}|/2)^{-C_{10}}}\,.$$

Taking logarithms gives $\log|\mathcal{I}| - (|\mathcal{I}|/2)^{-C_{10}} \log|\lambda| \leq \log(4C_7)$, which cannot hold when $|\mathcal{I}|$ is small enough. Hence b satisfies the theorem. □

Lemma 3.11. *Suppose $T, t_2, \ldots, t_\ell$, $\mathcal{I}, D, b_2, \ldots, b_\ell$, $1 < \delta_2 < \cdots < \delta_\ell < C_8$ satisfy the assertions (a-d) of Lem. 3.9. Then for every sufficiently small $\varepsilon > 0$ there exist a subinterval $\mathcal{I}'$ of $\mathcal{I}$ of length ε and $D' > \varepsilon^{-1/2}$ such that the assertions of Lem. 3.9 hold for $T, t_2, \ldots, t_\ell$, $\mathcal{I}', D', b_2, \ldots, b_\ell, \delta_2, \ldots, \delta_\ell$.*

Proof. For an interval $\mathcal{I}' \subset \mathbb{R}/(2\pi\mathbb{Z})$, let $D(\mathcal{I}')$ denote the smallest $d > 0$ such that $d \neq b_2, \ldots, b_\ell$ and $\mathcal{J}_d(1, \delta_\ell)$ and $\mathcal{I}'$ have a non-empty intersection. Set

$$c_1 = |\mathcal{I}| - \sum_{\substack{d=D \\ d \notin \{b_2,\ldots,b_\ell\}}}^{\infty} |\mathcal{J}_d(1, \delta_\ell)| \,.$$

By the assumption that assertion (d) of Lem. 3.9 is satisfied, $c_1 > 0$. Let $k \in \mathbb{N}$ such that $\frac{c_1}{k+2} < \varepsilon \leq \frac{c_1}{k+1}$ (which always exists for small enough ε). We will study the set

$$X = \mathcal{I} \setminus \bigcup_{\substack{d=D \\ d \notin \{b_2,\ldots,b_\ell\}}}^{k} \mathcal{J}_d(1, \delta_\ell) \,,$$

which is an interval $\mathcal{I}$ from which at most k intervals are removed. Thus, X consists of at most $k+1$ intervals. By assumption, $|X| > c_1$, and so $|X|$ contains an interval $\mathcal{I}'$ of length $\varepsilon \leq \frac{c_1}{k+1}$ such that $D(\mathcal{I}') \geq k+1$. Then, for small enough ε, we can take $D' = D(\mathcal{I}')$ as we then have that

$$D(\mathcal{I}') \geq k+1 \geq c_1/\varepsilon > \varepsilon^{-1/2} \,.$$

The interval $\mathcal{I}'$ inherits conditions (a) and (b) of Lemma 3.9 from $\mathcal{I}$ and condition (c) holds for D' by construction. For condition (d), write $\Delta = 1$ when $b < 1$ and $\Delta = 1/2$ when $b = 1$. By Lemmas 3.7 and 3.8, we have that for a constant c_2 and small enough ε, we have that

$$\sum_{d=D'}^{\infty} |\mathcal{J}_d(1, \delta_\ell)| \leq \sum_{d=D'}^{\infty} C_5 \frac{\delta_\ell - 1}{|\lambda|^{\Delta d}} = c_2 |\lambda|^{-\Delta D'} \leq c_2 |\lambda|^{-\Delta \varepsilon^{-1/2}} < \varepsilon = |\mathcal{I}'| \,. \qquad \square$$

Lemma 3.12. *Let $\mathcal{I} \subset \mathbb{R}/(2\pi\mathbb{Z})$ be an interval. Then there are an interval $\mathcal{I}'$ and constants $C_8 \geq 1, C_9$ with the following properties. Suppose $d_1, d_2 \geq C_9$ are such that the families of intervals $\mathcal{J}_{d_1}$ and $\mathcal{J}_{d_2}$ share a common anchor point that lies in $\mathcal{I}'$, i.e., $\arg(\chi_1 e^{-id_1\theta}) = \arg(\chi_2 e^{-id_2\theta}) \in \mathcal{I}'$, where each χ_1, χ_2 is one of $-b \pm \sqrt{b^2-1}$. Then $\mathcal{J}_{d_1}(1, C_8) \cap \mathcal{J}_{d_2}(1, C_8) = \emptyset$.*

Proof. First, suppose that $r^{d_1} = r^{d_2}$. Then, $\chi_1 = -br^{d_1} \pm_1 \sqrt{b^2-1}$ and $\chi_2 = -br^{d_2} \pm_2 \sqrt{b^2-1}$, where we adorned the two possible choices for the $\pm$-sign with indices to indicate their differences. If $\pm_1 = \pm_2$, we have that $\chi_1 = \chi_2$ and so $e^{id_1\theta} = e^{id_2\theta}$, and thus $e^{i\theta}$ is a root of unity, contradicting non-degeneracy. Next, if $b = 1$ then there is only one choice for $\chi_\pm$, and so

we can assume that $\pm_1 = \pm_2$. Moreover, in that case, $r = 1$ and so $r^{d_1} = r^{d_2}$. Thus, from now on, we can assume that $|b| < 1$.

Now assume that $r^{d_1} = r^{d_2}$ and $\pm_1 \neq \pm_2$. Then $d_2 - d_1$ is uniquely determined as $\chi_1/\chi_2 = e^{i(d_2-d_1)\theta}$, and any other solution would contradict non-degeneracy. Then let $b = r^{d_1}$ and

$$f_d(x) = \frac{\cos(x + d\theta) + b}{\cos(x) + b}|\lambda|^d .$$

Thus, $x \in \mathcal{J}_d(1, C_8)$ if and only if $f_d(x) \in (1, C_8)$. Then, by refining $\mathcal{I}$, we can assume that $f_d(x)$ is bounded on $\mathcal{I}$. Then, for a subinterval $\mathcal{I}'$ of $\mathcal{I}$, there are c_1, c_1', c_2, c_2' such that $f_{d_1}'(x) \in (c_1, c_1')$ and $f_{d_2}'(x) \in (c_2, c_2')$ for all $x \in \mathcal{I}'$, d_1, d_2 such that $\arg(\chi_1) - d_1\theta = \arg(\chi_2) - d_2\theta \in \mathcal{I}'$. When $|\mathcal{I}'|$ decreases, $c_i' - c_i$ goes to 0. As $f_{d_1}(x) - f_{d_2}(x)$ is never locally constant, (c_1, c_1') and (c_2, c_2') are disjoint for an interval $\mathcal{I}' \subset \mathcal{I}$. Then taking d_1, d_2 large enough, we have that $\mathcal{J}_{d_i}(1, C_8) \subset \mathcal{I}'$. Then, if $x - (\arg(\chi_1) - d_1\theta) \in \mathcal{J}_{d_i}(1, C_8)$, $x \in (1/c_i', C_8/c_i)$ and $(1/c_1', C_8/c_1)$ and $(1/c_2', C_8/c_2)$ are disjoint when $C_8 > 1$ is small enough.

If $r^{d_1} \neq r^{d_2}$, we have that $r^{d_1} = -r^{d_2}$. Then, $\chi_1 = -br^{d_1} \pm_1 \sqrt{b^2 - 1}$ and $\chi_2 = br^{d_2} \pm_2 \sqrt{b^2 - 1}$, where we again adorned the $\pm$-signs with indices. If $\pm_1 \neq \pm_2$, then $\chi_1 = -\chi_2$, and so $e^{id_1\theta} = -e^{id_2\theta}$, giving again that $e^{i\theta}$ is a root of unity, contradicting non-degeneracy. If $\pm_1 = \pm_2$, we end up with the same argument as in the case where $r^{d_1} = r^{d_2}$ and $\pm_1 \neq \pm_2$. □

Proof of Lem. 3.9. We inductively construct intervals $\mathcal{I}_1, \ldots, \mathcal{I}_\ell$ and integers $D_1, \ldots, D_\ell$ such that the result holds $\mathcal{I}_k$ and D_k when restricted to only these intervals while keeping the bound $(1, \delta_\ell)$ in items (c) and (d). Then we take $\mathcal{I} = \mathcal{I}_\ell$. As we will make sure that $\mathcal{I}_k \subset \mathcal{I}_{k+1}$, we can always use the same constants $C_5, \ldots, C_9$ from Lems. 3.7 and 3.8 and that $D_k \geq D_1$.

Base case ($k = 1$). As T is even and $n \equiv t_1 \pmod{T}$, it follows that $r^n = r^{t_1}$, and so set $b = r^{t_1}\beta$. Further let $\mathcal{I}_1 \subset \mathbb{R}/(2\pi\mathbb{Z})$, and ε be such that $\cos(x) + b > \varepsilon$ for all $x \in \mathcal{I}_1$. This is possible because either $|b| < 1$ or $b = r = 1$. Thus, (a) and (b) are satisfied.

Now maybe taking a subinterval of $\mathcal{I}_1$, there is a $\varepsilon' > 0$ such that for every ξ in the finite set Ξ defined in Lems. 3.7, 3.8, and 3.12 is at least ε' apart from any point in $\mathcal{I}_1$. Thus, we can use these constants $C_5, \ldots, C_9$ in these lemmas. Moreover, by Lem. 3.12, we can assume that when $d_1, d_2 \geq C_9$ and the anchor points $\mathcal{J}_{d_1}(1, C_8)$ and $\mathcal{J}_{d_2}(1, C_8)$ are equal, these intervals are disjoint. Now, as $\cos(x) + b > \varepsilon$ for all $x \in \mathcal{I}_1$, there is a $D' \in \mathbb{Z}$ such that $\cos(x) + b > \cos(x + \theta d)|\lambda|^d + br^d$ for all $d < D'$. Thus, there are only finitely

many $D' \leq d < C_9$. There are at most two x such that (4) holds. Taking $\mathcal{I}_1$ and $\delta_\ell > 1$ small enough, we can then guarantee that (c) also holds. Thus, we take $D_1 = C_9 > 0$. Let $\sigma(d) = 1$ if $|b| < 1$ and $\sigma(d) = 1/2$ if $b = 1$. Then using Lems.(3.7) and (3.8) and $|\lambda| > 1$, the geometric series

$$\sum_{d=D_1}^{\infty} \left|\mathcal{J}_d(1,\delta_\ell)\right| \leq (\delta_\ell - 1)C_6 \sum_{d=D_1}^{\infty} \lambda^{-\sigma(d)} = (\delta_\ell - 1)c$$

is finite. Thus, taking δ_ℓ close enough to 1 gives us the base case. For the other cases, choose $\delta_2, \ldots, \delta_{\ell-1}$ such that $\delta_1 := 1 < \delta_2 < \cdots < \delta_{\ell-1} < \delta_\ell$.

Induction step. Now let $k \geq 2$ and $\varepsilon > 0$. For ε small enough, applying Lem. 3.11 on $\mathcal{I}_{k-1}$, we get intervals $\mathcal{I}_\varepsilon \subset \mathcal{I}_{k-1}$ of length ε where the smallest d such that $\mathcal{J}_d(1,\delta_\ell) \cap \mathcal{I}_\varepsilon \neq \emptyset$ is at least $\varepsilon^{-1/2}$.

Lemma 3.10 implies that when ε is again small enough, there is a $\varepsilon^{-C_{10}} < b_k < 2\varepsilon^{-C_{10}}$ such that $b_k \equiv t_k \pmod{T}$ and $\mathcal{J}_{b_k}(\delta_{k-1},\delta_k) \subset \mathcal{I}_\varepsilon$.

If $d \notin \{0, b_2, \ldots, b_k\}$ and $\mathcal{J}_{b_k}(\delta_{k-1},\delta_k) \cap \mathcal{J}_d(1,\delta_\ell) \neq \emptyset$, then $d > \varepsilon^{-1/2}$. By Lem. 3.7, and because the anchor points $\chi_\pm e^{-ib_k\theta}$ and $\chi'_\pm e^{-id\theta}$ differ, we get

$$\left|(b_k - d)\theta - \arg(\chi_\pm) + \arg(\chi'_\pm)\right|_{2\pi} \leq \frac{C_7}{|\lambda|^{b_k}} + \frac{C_7}{|\lambda|^{d}} \leq \frac{2C_7}{|\lambda|^{\min\{b_k,d\}}}\,.$$

Then Baker's theorem (Thm. 2.12) gives a constant c_1 such that

$$|b_k - d|^{-c_1} < \left|(e^{i\theta})^{b_k-d}\frac{\chi_\pm}{\chi'_\pm} - 1\right| \leq |(b_k-d)\theta + \arg(\chi_\pm/\chi'_\pm)|_{2\pi} \leq \frac{2C_7}{|\lambda|^{\min\{b_k,d\}}}\,.$$

Taking logarithms, we have:

$$\min\{b_k, d\}\log|\lambda| < \log(2C_7) + c_1\log|b_k - d|\,. \tag{14}$$

Thus, if $d < b_k$, we have $d\log|\lambda| < \log(2C_7) + c_1\log(b_k)$. Using that $d > \varepsilon^{-1/2}$ and $b_k \leq 2(\varepsilon/2)^{-C_{10}}$, we obtain

$$\varepsilon^{-1/2}\log|\lambda| < \log(2C_7) + c_1\log(2) - c_1C_{10}\log(\varepsilon/2)\,.$$

This is impossible for sufficiently small ε. We can thus assume that $d > b_k$. We take $\mathcal{I}_k := \mathcal{J}_{b_k}(\delta_{k-1},\delta_k)$ such that items (a) and (b) are satisfied. Now assume $d > b_k$ and let $D = d$ (and so item (c) automatically holds). For

a contradiction, assume item (d) is violated. Lemma 3.7 and the geometric series give that for some constant $c_2 > 0$,

$$\frac{C_5(\delta_k - \delta_{k-1})}{|\lambda|^{b_k}} \leq |\mathcal{I}_k| \leq \sum_{d'=d}^{\infty} |\mathcal{J}_{d'}(1, \delta_\ell)| < c_2 |\lambda|^{-d} .$$

Hence, taking logarithms,

$$(d - b_k) \log |\lambda| \leq \log \left(\frac{c_2}{C_5(\delta_k - \delta_{k-1})} \right) .$$

which upper bounds $d - b_k$ and thus b_k due to (14). Induction completes the proof. □

Having established Lem. 3.9, we can prove Thm. 1.5, which finalizes our proof of Thm. 1.1: the MSO theory of the structure $\langle \mathbb{N}; <, \{u_n \colon n \in \mathbb{N}\} \cap \mathbb{N} \rangle$ is decidable when $\boldsymbol{u}$ is a simple, non-degenerate LRS with at most three dominant roots. The proof shares many ideas with the proof of Lem. 3.9.

Proof of Thm. 1.5. By Lem. 3.5, Thm. 1.5 follows when for all even $T \geq 2$ and $t_1, \ldots, t_\ell \in \{0, \ldots, T-1\}$, there are $b_2, \ldots, b_\ell \in \mathbb{N}$ such that $b_j \equiv t_j \pmod{T}$ and for all $N \in \mathbb{N}$ there is an $n \geq N$ congruent to t_1 modulo T and

1. $0 < v_n < v_{n+b_2} < \cdots < v_{n_\ell}$; and
2. when $v_n < v_m < v_{n+d_\ell}$ we have that $m - n \in \{b_2, \ldots, b_{\ell-1}\}$.

We will prove this statement for given N, T, ℓ, and $t_1, \ldots, t_\ell$. Lem. 3.9 gives particular numbers $b_2, \ldots, b_\ell \in \mathbb{N}$, $1 < \delta_\ell < C_8$ and an interval $\mathcal{I}$. Setting $n_j = n + b_j$ for $j = 2, \ldots, \ell$, we automatically have that $b_j \equiv t_j \pmod{T}$ when $n \equiv t_1 \pmod{T}$, and the first condition above is also satisfied. Thus, condition 2, $n \geq N$ and $n \equiv t_1 \pmod{T}$ remain.

Let $0 < \varepsilon < |\mathcal{I}|$ be small enough. Lemma 3.11 implies that there is an interval $\mathcal{I}_\varepsilon \subset \mathcal{I}$ of length ε such that $\mathcal{I}_\varepsilon \cap \mathcal{J}_d(1, \delta_\ell) = \emptyset$ for all $d < \varepsilon^{-1/2}$. By Lem. 3.10, there is a $\varepsilon^{-C_{10}} < n < 2\varepsilon^{-C_{10}}$ such that $n\theta + \varphi \in \mathcal{I}_\varepsilon$ and $n \equiv t_1 \pmod{T}$. Thus, $n \geq N$ for small enough ε, and only condition 2 remains.

Now assume that $m - n \notin \{b_2, \ldots, b_{\ell-1}\}$ and $v_n < v_m < v_{n+b_\ell}$. Then, setting $d = m - n$, we have that $n\theta + \varphi \in \mathcal{J}_d(1, \delta_\ell)$ and thus $d \geq \varepsilon^{-1/2}$.

If $n\theta + \varphi$ would be equal to one of the anchor points $\arg(\chi_\pm) - d\theta$ of $\mathcal{J}_d(1, C_8)$, then $n\theta + \varphi \notin \mathcal{J}_d(1, C_8)$. Otherwise, an interval of $\mathcal{J}_d(0, 1)$ has

length zero, which is absurd. Thus, $n\theta + \varphi \neq \arg(\chi_\pm) - d\theta$, and Baker's theorem (Thm. 2.12) gives a constant c_1 such that

$$\begin{aligned} C_7|\lambda|^{-d} &\geq |(n\theta + \varphi) - (\arg(\chi_\pm) - d\theta)|_{\pi/2} \\ &\geq |e^{i\theta(n+d)}e^{i\varphi}\chi_\pm - 1| \\ &> |n+d|^{-c_1}\,. \end{aligned}$$

Taking logarithms, we obtain $\log(C_7) + c_1\log(n+d) > d\log|\lambda|$. As $d \geq \varepsilon^{-1/2}$ and $n \geq \varepsilon^{-C_{10}}$, we have that $n, d \geq 2$ for small enough ε. Then, $dn \geq n + d$ and so

$$\log(C_7) + c_1\log(n) + c_1\log(d) > d\log|\lambda|\,.$$

Hence, either

$$2\log(C_7) + 2c_1\log(d) > d\log|\lambda| \quad \text{or} \quad 2c_1\log(n) > d\log|\lambda|\,.$$

The former is impossible for large enough d (and thus small enough ε), while for the latter, the upper and lower bounds for n and d give that

$$2c_1\log(2\varepsilon^{-C_{10}}) > 2c_1\log(n) > d\log|\lambda| > \log|\lambda|\varepsilon^{-1/2}\,,$$

which again is impossible for small enough ε. Condition 2 follows. □

4. Concluding Remarks

Our main result, Thm. 1.1, significantly expands the decidability landscape of MSO theories of structures of the form $\langle \mathbb{N}; <, P\rangle$, by considering predicates P obtained as sets of positive terms of non-degenerate simple LRS having at most three dominant roots. The following strengthening of Thm. 1.1 can be deduced quickly from our proof: given an LRS $\langle u_n\rangle_{n=0}^\infty$ as in the statement, and an MSO formula φ, it is decidable whether φ holds in the structure $\langle \mathbb{N}; <, \{u_n \colon n \in \mathbb{N}\} \cap \mathbb{N}\rangle$. The formulation of Thm. 1.1 fixes P and, technically speaking, allows for a non-uniform family of algorithms, e.g., one algorithm per structure. However, we do not use any non-uniform arguments in our proof.

A natural question is whether any of these constraints in the statement of Thm. 1.1 can be relaxed. Let us begin with the number of dominant roots. The famously open Skolem (resp. Positivity) problem asks to decide, given an integer-valued LRS $\langle u_n\rangle_{n=0}^\infty$, decide whether $u_n = 0$ (resp. $u_n > 0$) for some n;

see, e.g. [30]. The Skolem problem has remained open (for at least 40, and by some measure, 100 years) for simple, non-degenerate LRS with exactly 4 dominant roots; hence a *uniform* generalisation of Thm. 1.1 to 4 or more dominant roots would entail major mathematical breakthroughs. A non-uniform generalisation (i.e., where the algorithm is allowed access to some finite, mathematically intricate information about P) seems equally unlikely to us: for LRS $\langle u_n \rangle_{n=0}^{\infty}$ with 4 or dominant roots, no general approach is known to even determining whether given n belongs to P.

On the other hand, deciding whether a given non-simple LRS with three dominant roots has finitely many positive terms is *Diophantine-hard*: A solution would solve well-established open problems in number theory [31]. Finally, we mention that the restriction that $\langle u_n \rangle_{n=0}^{\infty}$ be non-degenerate is essential: lifting it has the effect of voiding the constraint on the number of dominant roots.[4] Even more problematic is the fact that, one can interleave non-degenerate LRS into a single, possibly degenerate LRS: if $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(k)}$ are LRS with value sets $P_i = \{u_n^{(i)} \colon n \in \mathbb{N}\}$, then there is a possibly degenerate LRS $\boldsymbol{u}$ with value set P such that $P = \bigcup_{i=1}^{k} P_i$. Hence, studying the MSO theory of $\langle \mathbb{N}; < \rangle$ expanded with the value set of a single possibly degenerate LRS is as difficult as studying MSO theories of expansions with finitely many LRS.

Although further decidability results are currently out of reach, we conjecture that our technique, namely using effective almost periodicity and prodisjunctivity, is sufficient for a large class of LRS. As Ouaknine and Worrell [32] already classified which LRS have finitely many positive terms, we only give our conjecture to split the almost periodic and disjunctive cases.

Conjecture 4.1. *Let $\boldsymbol{u}$ be a simple, non-degenerate LRS such that $P = \{u_n \colon n \in \mathbb{N}\}$ is infinite. Enumerate P as $\langle p_m \rangle_{m=0}^{\infty}$. Then $\langle p_m \bmod M \rangle_{m=0}^{\infty}$*

- *is almost periodic for all $M \geq 1$ if there is a $D \in \mathbb{N}$ such that $u_n > u_m$ whenever $n > m + d$ and $u_m \geq 0$;*
- *is disjunctive otherwise.*

[4] More precisely, one can construct a degenerate LRS for which the two dominant roots "cancel" each other out at every odd index, and the sub-LRS that remains over odd indices (involving the non-dominant roots) can be arbitrarily complex.

References

[1] J. R. Büchi, Symposium on decision problems: On a decision method in restricted second order arithmetic, in: Studies in Logic and the Foundations of Mathematics, Vol. 44, Elsevier, 1966, pp. 1–11.

[2] C. C. Elgot, M. O. Rabin, Decidability and undecidability of extensions of second (first) order theory of (generalized) successor, The Journal of Symbolic Logic 31 (2) (1966) 169–181.

[3] O. Carton, W. Thomas, The Monadic Theory of Morphic Infinite Words and Generalizations, Information and Computation 176 (1) (2002) 51–65.

[4] A. Rabinovich, On decidability of monadic logic of order over the naturals extended by monadic predicates, Information and Computation 205 (6) (2007) 870–889.

[5] A. Rabinovich, W. Thomas, Decidable theories of the ordering of natural numbers with unary predicates, in: International Workshop on Computer Science Logic, Springer, 2006, pp. 562–574.

[6] V. Berthé, T. Karimov, M. Vahanwala, Automata on S-adic words, arXiv preprint arXiv:2506.17460 (2025).

[7] A. L. Semenov, Logical theories of one-place functions on the set of natural numbers, Mathematics of the USSR-Izvestiya 22 (3) (1984) 587.

[8] A. Semenov, Decidability of monadic theories, in: International Symposium on Mathematical Foundations of Computer Science, Springer, 1984, pp. 162–175.

[9] P. Bateman, C. Jockusch, A. Woods, Decidability and undecidability of theories with a predicate for the primes, The Journal of symbolic logic 58 (2) (1993) 672–687.

[10] T. Karimov, E. Kelmendi, J. Ouaknine, J. Worrell, What's Decidable About Discrete Linear Dynamical Systems?, in: Principles of Systems Design, Vol. 13660 of Lecture Notes in Computer Science, Springer, 2022, pp. 21–38.

[11] V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, J. Worrell, On the Decidability of Monadic Second-Order Logic with Arithmetic predicates, in: LICS, ACM, 2024, pp. 11:1–11:14.

[12] W. Thomas, A note on undecidable extensions of monadic second order successor arithmetic, Archiv für mathematische Logik und Grundlagenforschung 17 (1) (1975) 43–44.

[13] V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, J. Worrell, On the decidability of monadic theories of arithmetic predicates, arXiv preprint arXiv:arXiv:2405.07953 (2025).

[14] S. Shelah, The monadic theory of order, Ann. Math. 102 (1975) 379–419.

[15] W. Thomas, Ehrenfeucht Games, the Composition Method, and the Monadic Theory of Ordinal Words, in: Structures in Logic and Computer Science, Vol. 1261 of Lecture Notes in Computer Science, Springer, 1997, pp. 118–143.

[16] J. Nieuwveld, J. Ouaknine, On expansions of monadic second-order logic with dynamical predicates, in: P. Gawrychowski, F. Mazowiecki, M. Skrzypczak (Eds.), 50th International Symposium on Mathematical Foundations of Computer Science, MFCS 2025, Warsaw, Poland, August 25-29, 2025, Vol. 345 of LIPIcs, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, pp. 80:1–80:17. doi:10.4230/LIPICS.MFCS.2025.80.

[17] É. M. Borel, Les probabilités dénombrables et leurs applications arithmétiques, Rendiconti del Circolo Matematico di Palermo (1884-1940) 27 (1) (1909) 247–271.

[18] G. Harman, One hundred years of normal numbers, in: Surveys in Number Theory, AK Peters/CRC Press, 2002, pp. 57–74.

[19] M. Queffélec, Old and new results on normality, Lecture Notes-Monograph Series (2006) 225–236.

[20] L. Kuipers, H. Niederreiter, Uniform distribution of sequences, Courier Corporation, 2012.

[21] Y. Bugeaud, Distribution modulo one and Diophantine approximation, Vol. 193, Cambridge University Press, 2012.

[22] A. Dubickas, M. Sha, Positive density of integer polynomials with some prescribed properties, Journal of Number Theory 159 (2016) 27–44.

[23] V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, J. Worrell, The monadic theory of toric words, Theoretical Computer Science 1025 (2025) 114959.

[24] A. Blumensath, Monadic Second-Order Model Theory, `http://www.fi.muni.cz/~blumens/`, [Online; accessed on 24 January 2025] (2024).

[25] W. Thomas, Languages, automata, and logic, in: Handbook of Formal Languages: Volume 3 Beyond Words, Springer, 1997, pp. 389–455.

[26] S. M. Gonek, H. L. Montgomery, Kronecker's approximation theorem, Indagationes Mathematicae 27 (2) (2016) 506–523, in Memoriam J.G. Van der Corput (1890–1975) Part 2.

[27] A. Baker, G. Wüstholz, Logarithmic forms and group varieties, Journal für die reine und angewandte Mathematik 442 (1993) 19–62.

[28] M. Mignotte, T. N. Shorey, R. Tijdeman, The distance between terms of an algebraic recurrence sequence., J. Reine Angew. Math. 346 (1984) 63–76.

[29] L. Kronecker, Zwei Sätze über Gleichungen mit ganzzahligen Coefficienten, J. Reine Angew. Math. 53 (1857) 173–175.

[30] G. Everest, A. van der Poorten, I. Shparlinski, T. Ward, Recurrence Sequences, Mathematical Surveys and Monographs, American Mathematical Society, 2003.

[31] J. Ouaknine, J. Worrell, Positivity Problems for Low-Order Linear Recurrence Sequences, in: SODA, SIAM, 2014, pp. 366–379.

[32] J. Ouaknine, J. Worrell, Ultimate positivity is decidable for simple linear recurrence sequences, in: International Colloquium on Automata, Languages, and Programming, Springer, 2014, pp. 330–341.